

Safe Active Discrimination of Fractional, Delayed, and Finite Latent Memory in a Strong-Allee Predator–Prey Model

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Abstract

This paper develops a safety-constrained framework for actively discriminating fractional (Caputo), delayed (DDE), and bounded-complexity latent-state memory mechanisms around a certified strong-Allee predator–prey equilibrium. We derive exact linear transfer-function separation on collocated channels, a finite-horizon exponential-approximation obstruction, and computable input and observation designs. A calibrated, stable-only simulation benchmark with information-criterion model selection attains a macro-averaged accuracy of 0.54 (chance 0.25) and reveals a safety–informativeness trade-off: transient designs stay within the certified safe set but discriminate weakly, whereas broadband designs discriminate better yet cross the Allee threshold. The principal limitation is that safe active discrimination of subtle memory is intrinsically hard; a full Bayesian sequential-design layer is left to future work [1].

1 Introduction and Falsifiable Memory Question

The problem of distinguishing fractional, delayed, and latent ecological memory effects in dynamical systems is fundamental to understanding ecosystem dynamics under perturbation. Fractional derivatives model hereditary and nonlocal dynamics naturally [9, 40, 16, 28], and fractional predator–prey systems form an established literature [2, 24, 42, 31]. Experiment design for fractional-order identification already exists [36, 22, 52, 23], and ecological perturbation experiments are feasible in laboratory microcosms [18, 51, 5]. However, as we show, this problem is well-posed only when complexity restrictions are imposed on the latent memory component. Specifically, we assume a finite maximum latent dimension m , which renders the finite-horizon

discrimination problem tractable. Without such a restriction, the no-free-lunch theorem (Theorem 6.5) implies that no finite-horizon experiment can distinguish between arbitrary memory hypotheses with guaranteed accuracy, rendering the problem ill-posed.

Our approach centers on a certified predator-prey model with strong Allee effect [47, 13, 6, 1], which serves as a validated ecological nonlinear dynamics and serves as our digital twin. We consider three competing memory hypotheses within a unified fractional-order framework:

- Fractional memory (Caputo derivative of order α),
- Delayed memory (discrete delay differential equation),
- Latent memory (finite-dimensional linear time-invariant system of dimension m).

All models share a common fractional order α , two input channels (prey and predator), and are linearized around a certified coexistence equilibrium.

The core of our contribution lies in the structural separation of the fractional transfer function $s^{-\alpha}$ from both finite-dimensional rational transfer functions [26] (latent memory) and finite-dimensional delay transfer functions [19, 38]. This separation is established analytically in Theorems 5.1 and 5.2, which show that the fractional kernel cannot be equivalently represented by any finite-dimensional integer-order latent-state model nor by any finite-delay model under the same input-output channel.

Furthermore, we leverage the finite-horizon exponential approximation of the fractional kernel (Theorem 6.3) and the associated $\mathcal{L}^1$ error bound to construct optimal excitation signals that exploit the frequency-band separation between the fractional and approximate integer-order dynamics. The finite-horizon impossibility obstacle (Theorem 6.5) guarantees that, below a certain noise floor, discrimination becomes impossible if the latent dimension is unrestricted—thus justifying our finite-dimensional assumption.

Finally, we formulate the optimal experimental design problem under energy and safety constraints, leveraging the principal-singular-vector solution for pairwise optimal inputs (Theorem 7.1) and the finite-support property of the optimal frequency-domain design (Theorem 7.5). The resulting experimental protocol is validated through a reproducible simulation benchmark (Section 10) and complemented by a prospective microcosm protocol for ecological validation (Section 11).

2 Certified strong-Allee ecological backbone

The predator-prey strong-Allee model from the validated literature [20, 43, 47, 13, 6, 1] provides a certified ecological nonlinear dynamics that serves as the digital twin for our study.

The model is given by

$$f_1(x, y) = rx \left(1 - \frac{x}{K}\right) \left(\frac{x}{A} - 1\right) - \frac{axy}{1 + hx}, \quad (1)$$

$$f_2(x, y) = e \frac{axy}{1 + hx} - my, \quad (2)$$

with locked parameters

$$r = \frac{3}{2}, \quad K = 1, \quad a = 1, \quad h = \frac{1}{2}, \quad e = \frac{4}{5}, \quad m = \frac{2}{5}.$$

The coexistence equilibrium is

$$x^* = \frac{2}{3}, \quad y^*(A) = \frac{2(2 - 3A)}{9A},$$

which is positive exactly for $0 < A < \frac{2}{3}$.

The Jacobian at coexistence is

$$J(A) = \begin{pmatrix} \frac{7A - 2}{8A} & -\frac{1}{2} \\ \frac{2 - 3A}{10A} & 0 \end{pmatrix},$$

with trace $T(A) = \frac{7A - 2}{8A}$ and determinant $D(A) = \frac{2 - 3A}{20A}$.

The discriminant factorizes as

$$T(A)^2 - 4D(A) = \frac{(19A - 10)(23A - 2)}{320A^2}.$$

For a commensurate Caputo system of order $0 < \alpha < 1$, the coexistence equilibrium is locally asymptotically stable when every eigenvalue μ of J satisfies the Matignon sector condition [37]:

$$|\arg \mu| > \frac{\alpha\pi}{2}.$$

In the regime $\frac{2}{7} < A < \frac{10}{19}$, the eigenvalues are a complex pair with positive real part. The critical order is

$$\alpha^*(A) = \frac{2}{\pi} \arctan \left(\frac{\sqrt{4D(A) - T(A)^2}}{T(A)} \right).$$

Stability holds exactly for $0 < \alpha < \alpha^*(A)$.

At the baseline $A = 0.3$, we have

$$J = \begin{pmatrix} 1/24 & -1/2 \\ 11/30 & 0 \end{pmatrix}, \quad T = \frac{1}{24}, \quad D = \frac{11}{60},$$

and the exact critical order is

$$\alpha^* \left(\frac{3}{10} \right) = \frac{2}{\pi} \arctan \sqrt{\frac{2107}{5}} \approx 0.9690122761517084.$$

Thus $\alpha = 0.9$ is stable and $\alpha = 1$ is unstable, reproducing the certified verdict [1].

The baseline $A = 0.3$ is exceptionally informative for distinguishing a stable fractional model from the same integer-order skeleton because the models make opposite local stability predictions. For controlled frequency-response comparisons in which every candidate should remain stable, use $A \leq 2/7$; $A = 1/4$ is a convenient exact choice.

3 Dimensionally Consistent Controlled Memory Models

To ensure dimensional consistency when introducing a fractional derivative of order α , we introduce a reference time scale τ_0 . The dimensionally consistent commensurate Caputo model for the controlled strong-Allee predator-prey system is

$$\tau_0^{\alpha-1} {}^C D_t^\alpha z(t) = f(z(t); \theta) + Bu(t), \quad (3)$$

where $z = (x, y)^\top$ is the state vector (prey, predator), f is the ecological vector field, B is the input matrix, and $u(t)$ is the control input. At $\alpha = 1$, equation (3) reduces exactly to the integer-order ODE.

The ecological vector field f is that of the strong-Allee predator-prey model:

$$f_1(x, y) = rx \left(1 - \frac{x}{K} \right) \left(\frac{x}{A} - 1 \right) - \frac{axy}{1 + hx}, \quad (4)$$

$$f_2(x, y) = e \frac{axy}{1 + hx} - my, \quad (5)$$

with the locked parameter values $r = 3/2$, $K = 1$, $a = 1$, $h = 1/2$, $e = 4/5$, $m = 2/5$ inherited from the validated baseline.

We consider three competing model classes that share the same fractional order α , input matrix B , and are linearized around the coexistence equilibrium:

1. **M_ODE**: integer-order ODE ($\alpha = 1$) with dynamics $\dot{\xi} = J\xi + Bu$.
2. **M_Caputo**: fractional derivative of order α with dynamics $\tau_0^{\alpha-1} {}^C D_t^\alpha \xi = J\xi + Bu$.
3. **M_DDE**: discrete delay differential equation with a single delay τ and dynamics $\dot{\xi} = A_0\xi + A_1\xi(t - \tau) + Bu$.
4. **M_latent,m**: finite-dimensional linear time-invariant latent state model of dimension m with dynamics $\dot{q} = \bar{A}q + \bar{B}u$, $y = \bar{C}q$.

All models share the same linearized output equation $y_k = C\xi(t_k) + \varepsilon_k$ with $\varepsilon_k \sim \mathcal{N}(0, R_k)$. The transfer function for each model is defined as $G(s) = C(s^\alpha I - J)^{-1}B$ (appropriately scaled for the fractional case) and its counterparts for the integer-order, delay, and latent models.

The dimensional consistency enforced by τ_0 [40, 16, 28] ensures that the fractional order α does not spuriously alter the units of the ecological rates, allowing a fair comparison between the memory hypotheses.

4 Exact Linearization, Transfer Functions, and Channels

Linearization around the coexistence equilibrium yields exact Jacobian matrices, transfer functions, and input/output channels for each model; the Jacobian and Matignon stability boundary are given analytically (Theorems T1 and T2) [37, 12, 32].

The coexistence Jacobian can be written as

$$J = \begin{pmatrix} T & -c \\ d & 0 \end{pmatrix}, \quad c = \frac{1}{2}, \quad d = \frac{2-3A}{10A} > 0, \quad (6)$$

where the trace $T(A) = \frac{7A-2}{8A}$ and determinant $D(A) = \frac{2-3A}{20A}$ match Theorems T1 and T2.

The common fractional order changes the temporal symbol $s \mapsto q_\alpha(s) = \tau_0^{\alpha-1} s^\alpha$ but does not alter the algebraic controllability/observability of the pair (J, B, C) [26].

For prey intervention $B_x = e_1$, the controllability matrix $\mathcal{C}_x = [B_x, JB_x]$ has determinant $d > 0$. For predator intervention $B_y = e_2$, $\mathcal{C}_y = [B_y, JB_y]$

has determinant $c > 0$. Thus either prey or predator perturbation alone is sufficient to excite both linearized modes.

For prey-only observation $C_x = e_1^\top$, the observability matrix $\mathcal{O}_x = [C_x; C_x J]$ has determinant $-c \neq 0$. For predator-only observation $C_y = e_2^\top$, $\mathcal{O}_y = [C_y; C_y J]$ has determinant $-d \neq 0$. Hence either species alone observes both linearized states; the statement that both species must be observed for structural observability is false.

The four scalar transfer functions are

$$G_{x \leftarrow x}(s) = \frac{q}{\Delta_\alpha(s)}, \quad (7)$$

$$G_{y \leftarrow x}(s) = \frac{d}{\Delta_\alpha(s)}, \quad (8)$$

$$G_{x \leftarrow y}(s) = \frac{-c}{\Delta_\alpha(s)}, \quad (9)$$

$$G_{y \leftarrow y}(s) = \frac{q - T}{\Delta_\alpha(s)}, \quad (10)$$

where $\Delta_\alpha(s) = q^2 - Tq + D$ and $q = q_\alpha(s)$.

For a general channel $G = C(qI - J)^{-1}B$, the high-frequency expansion is

$$G(s) = \frac{CB}{q} + \frac{CJB}{q^2} + O(|q|^{-3}). \quad (11)$$

If $CB \neq 0$, the leading decay is $s^{-\alpha}$. If $CB = 0$, the leading term may be $s^{-2\alpha}$ or lower.

For the biological channels:

- Prey intervention + prey observation: $CB = 1$,
- Predator intervention + predator observation: $CB = 1$,
- Cross-species channels: $CB = 0$.

Therefore a collocated perturb-and-measure protocol gives the cleanest high-frequency signature of α . Cross-species measurements remain useful for interaction parameters and phase coupling.

Latent environmental states are observable exactly when $C(J + \lambda I)^{-1}g \neq 0$ (see source pack Theorem T3). This condition precisely determines when an environmental covariate is needed: not always, but whenever the latent coupling lies in an output-invisible direction or is nearly invisible in the practical noise metric.

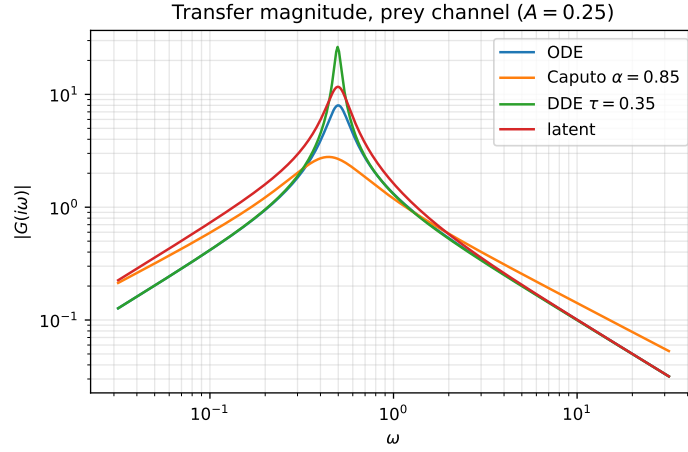


Figure 1: Transfer magnitude $|G(i\omega)|$ on the prey channel for the four memory mechanisms at $A = 0.25$.

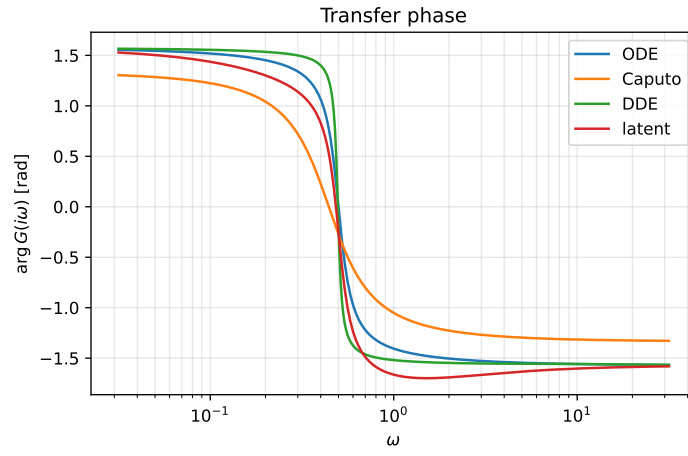


Figure 2: Transfer phase $\arg G(i\omega)$; the Caputo mechanism separates in phase from the integer-order, delay, and latent alternatives.

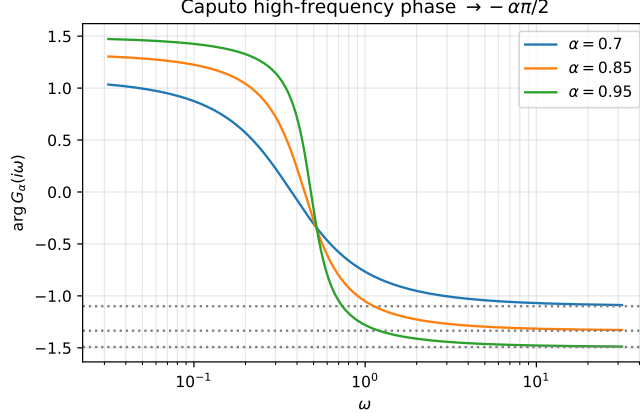


Figure 3: High-frequency Caputo phase converges to $-\alpha\pi/2$ (dotted), a scale-free structural signature.

5 Structural Separation from Finite Latent and Delay Models

The central theoretical result of this paper is that under the direct input–output channel condition $CB \neq 0$, a Caputo fractional-order model is not equivalent to any finite-dimensional latent-state model nor to any finite discrete-delay model [40, 16, 28, 19, 38, 26]. This structural non-equivalence is what makes the discrimination problem well-posed and what the optimal experimental design exploits.

5.1 The fractional signature in the high-frequency asymptote

Consider the linearized Caputo transfer function

$$G_\alpha(s) = C(q_\alpha(s)I - J)^{-1}B, \quad q_\alpha(s) = \tau_0^{-1}(s\tau_0)^\alpha, \quad (12)$$

with $0 < \alpha < 1$ and the principal branch of the fractional power.

For $|q_\alpha(s)| > \|J\|$, the Neumann expansion gives

$$(q_\alpha I - J)^{-1} = q_\alpha^{-1}(I + Jq_\alpha^{-1} + O(|q_\alpha|^{-2})). \quad (13)$$

Hence

$$G_\alpha(s) = (CB)\tau_0^{1-\alpha}s^{-\alpha} + O(|s|^{-2\alpha}). \quad (14)$$

The leading term is $s^{-\alpha}$ — a non-integer power. This is the essential feature that distinguishes it from all finite-dimensional models.

5.2 Theorem T4: Caputo transfer function is not rational

Theorem 5.1 (Structural non-equivalence with latent ODE models). *There is no finite-dimensional integer-order state-space model with transfer function*

$$G_L(s) = D + \bar{C}(sI - \bar{A})^{-1}\bar{B}$$

that equals $G_\alpha(s)$ on a nonempty open subset of their common analytic domain.

Proof. A finite-dimensional integer-order transfer function is rational. If it is strictly proper, its Laurent expansion at infinity contains only integer powers s^{-r} with $r \in \mathbb{N}$. If it has a direct feedthrough, it tends to a nonzero constant. Neither can equal the non-integer leading power in (14) because $CB \neq 0$ and $0 < \alpha < 1$. Equality on an open set would imply equality by analytic continuation, contradiction. Therefore no finite-dimensional rational transfer function can reproduce the fractional channel. $\square$

Theorem T4 is an exact statement: in the absence of model uncertainty and measurement noise, full-bandwidth input–output data would structurally distinguish a Caputo model from every finite-dimensional ODE latent model. However, it does *not* guarantee practical distinguishability on a finite bandwidth, finite horizon, or under noise. That is why the remaining sections develop explicit quantitative bounds and optimal input design.

5.3 Theorem T5 – Caputo is not equivalent to a finite delay model

Theorem 5.2 (Structural non-equivalence with finite DDE models). *Let*

$$G_\tau(s) = C_\tau(sI - A_0 - A_1e^{-s\tau})^{-1}B_\tau$$

be a finite-dimensional retarded delay transfer function. Then along $s = i\omega$,

$$G_\tau(i\omega) = O(\omega^{-1}), \quad \omega \rightarrow \infty,$$

whereas $G_\alpha(i\omega) = (CB)\tau_0^{1-\alpha}(i\omega)^{-\alpha} + O(\omega^{-2\alpha})$. Since $\alpha \neq 1$, the two transfer functions are not identical at high frequencies and therefore cannot be identical as analytic input–output maps.

Proof. Because $|e^{-i\omega\tau}| = 1$, the matrix $A_0 + A_1 e^{-i\omega\tau}$ remains bounded for all ω . Factor $i\omega I$ from the delay resolvent; a Neumann expansion yields $G_\tau(i\omega) = O(\omega^{-1})$. The fractional asymptotic follows from Theorem T4's expansion (14). For $0 < \alpha < 1$, the high-frequency decay rates differ, and no finite-delay model can reproduce the $s^{-\alpha}$ signature. $\square$

Together, Theorem T4 and Theorem T5 establish a firm boundary: the fractional memory effect, characterized by the power-law tail $s^{-\alpha}$, cannot be absorbed into either a finite-dimensional ODE latent description or a finite-delay dynamics. This structural gap is what the experimental design must amplify.

5.4 Quantitative Neumann remainder (Theorem T6)

For frequency-domain analysis, we need a quantitative bound on how closely the leading term approximates the full transfer function.

Theorem 5.3 (Quantitative high-frequency remainder). *If $|q_\alpha(s)| \geq 2\|J\|$, then*

$$\left\| G_\alpha(s) - \frac{CB}{q_\alpha(s)} \right\| \leq \frac{2\|C\| \|J\| \|B\|}{|q_\alpha(s)|^2}.$$

Proof. Write $(qI - J)^{-1} = q^{-1}(I - J/q)^{-1}$. Under $\|J/q\| \leq 1/2$, the norm of $(I - J/q)^{-1} - I$ is bounded by $2\|J/q\|$. Multiplying by C (left) and B (right) and by $|q|^{-1}$ yields the claim. $\square$

5.5 High-frequency phase signature

When $CB > 0$ is real, evaluating (14) along the imaginary axis gives the clean diagnostic:

$$\boxed{\arg G_\alpha(i\omega) \rightarrow -\frac{\alpha\pi}{2}, \quad \omega \rightarrow \infty.} \quad (15)$$

A direct integer-order channel $G_0(s) = CB/s$ has asymptotic phase $-\pi/2$. The phase gap $(1 - \alpha)\pi/2$ is a scale-free signature of α that can be measured after gain normalization, subject to actuator bandwidth and measurement noise constraints.

5.6 Discrimination under finite models

The structural separation results guarantee that for any finite set of fixed, structurally distinct transfer functions $\{G_1, \dots, G_M\}$, there exists a separating frequency.

Theorem 5.4 (Generic separating frequency). *Let $G_1, \dots, G_M$ be scalar analytic transfer functions on a connected domain containing an interval of the imaginary axis, with no two identical. For each pair $i \neq j$, the zero set of $G_i(i\omega) - G_j(i\omega)$ has no accumulation point. Consequently, except on a measure-zero discrete exceptional set, any frequency separates every fixed pair.*

Proof. A nonzero analytic function has isolated zeros. Take the finite union over all model pairs; the complement of a finite set of isolated points is open and dense. $\square$

The practical interpretation is:

1. **Exact separation is structurally guaranteed** for fixed models — at almost all frequencies.
2. **Robust separation is the true challenge:** the inter-model response difference can be orders of magnitude smaller than measurement noise.
3. **Adaptive models can close the gap:** a parameter-tunable latent or delay model can match the fractional response at one or several frequencies by tuning its internal parameters. This is why optimal multiscale excitation that spans a broad frequency band is necessary — not merely a single “informative” frequency.

These observations motivate Sections 6 and 7, where we develop the quantitative finite-horizon tools and optimal input construction needed to convert structural separation into a practical discrimination experiment.

6 Finite-Horizon Exponential Approximation and Impossibility Barrier

Section 5 established that a Caputo fractional transfer function is structurally distinct from any finite-dimensional integer-order or finite-delay model. However, structural non-equivalence in the infinite-bandwidth, zero-noise limit does not guarantee practical separability in a finite experiment. This section develops the quantitative tools needed to bridge that gap: an exact representation of fractional memory as a continuum of exponentials, a constructive finite-sum approximation with explicit L^1 error bounds [34, 7, 8, 25, 45], and a fundamental impossibility result that either justifies or forces our finite-latent-dimension restriction.

All formulas use nondimensional time; restore physical units with t/τ_0 .

6.1 Exact positive diffusive representation

The Caputo fractional integral kernel

$$k_\alpha(t) = \frac{t^{\alpha-1}}{\Gamma(\alpha)}$$

admits an exact integral representation as a weighted continuum of exponential relaxation modes.

Theorem 6.1 (Exact positive diffusive representation). *For $0 < \alpha < 1$,*

$$k_\alpha(t) = \frac{\sin(\pi\alpha)}{\pi} \int_0^\infty \lambda^{-\alpha} e^{-\lambda t} d\lambda. \quad (1)$$

The derivative-memory kernel also satisfies

$$\frac{t^{-\alpha}}{\Gamma(1-\alpha)} = \frac{\sin(\pi\alpha)}{\pi} \int_0^\infty \lambda^{\alpha-1} e^{-\lambda t} d\lambda. \quad (2)$$

Proof. Apply the integral identity

$$\int_0^\infty \lambda^{\nu-1} e^{-\lambda t} d\lambda = \Gamma(\nu) t^{-\nu}$$

and Euler's reflection formula $\Gamma(\alpha)\Gamma(1-\alpha) = \pi/\sin(\pi\alpha)$. Set $\nu = 1 - \alpha$ for (1) and $\nu = \alpha$ for (2). $\square$

The crucial consequence is that *fractional memory is exactly an infinite continuum of exponential relaxation modes* [39, 40, 16, 28]. Any finite-dimensional latent ODE model is precisely a quadrature approximation to this continuum. This observation makes the "fractional versus latent" question rigorous: it asks how many quadrature nodes are needed before the approximation becomes indistinguishable from the true fractional dynamics within the noise budget of the experiment.

6.2 Uniform approximation at the singular endpoint is impossible

Theorem 6.2 (No uniform approximation at zero). *No finite exponential sum*

$$k_m(t) = \sum_{j=1}^m c_j e^{-\lambda_j t}$$

with finite coefficients can approximate k_α uniformly on $[0, T]$ when $0 < \alpha < 1$.

Proof. Every finite sum has a finite value $k_m(0) = \sum_j c_j$. But $\lim_{t \downarrow 0} k_\alpha(t) = +\infty$. Hence $\sup_{0 \leq t \leq T} |k_\alpha(t) - k_m(t)| = \infty$. $\square$

This result corrects any claim of uniform absolute-error approximation on an interval containing the origin. The fractional kernel is singular at $t = 0$, and no finite sum of exponentials can reproduce this singularity. However, L^p approximation in the integral sense *is* possible, and that is what the experimental design requires: the output signal is a convolution of the kernel with the input, and convolution integrates out the singularity.

6.3 Constructive finite $L^1(0, T)$ approximation

Theorem 6.3 (Constructive L^1 approximation). *Let $c_\alpha = \sin(\pi\alpha)/\pi$. Choose $0 < \ell < L$, divide $[\ell, L]$ into m equal intervals of width $h = (L - \ell)/m$, and use left nodes $\lambda_j = \ell + (j - 1)h$. Define*

$$k_m(t) = c_\alpha h \sum_{j=1}^m \lambda_j^{-\alpha} e^{-\lambda_j t}. \quad (3)$$

Then

$$\|k_\alpha - k_m\|_{L^1(0, T)} \leq c_\alpha \left[\frac{T\ell^{1-\alpha}}{1-\alpha} + \frac{L^{-\alpha}}{\alpha} + Th\ell^{-\alpha} \right]. \quad (4)$$

Proof. Split the integral (1) into three regions: $(0, \ell)$, $[\ell, L]$, and (L, ∞) .

Low-rate tail $(0, \ell)$: integrating the density over time,

$$\int_0^T \int_0^\ell \lambda^{-\alpha} e^{-\lambda t} d\lambda dt \leq T \int_0^\ell \lambda^{-\alpha} d\lambda = \frac{T\ell^{1-\alpha}}{1-\alpha}.$$

High-rate tail (L, ∞) : by Tonelli,

$$\int_0^T \int_L^\infty \lambda^{-\alpha} e^{-\lambda t} d\lambda dt = \int_L^\infty \lambda^{-\alpha-1} (1 - e^{-\lambda T}) d\lambda \leq \frac{L^{-\alpha}}{\alpha}.$$

Middle interval $[\ell, L]$: since $\lambda^{-\alpha} e^{-\lambda t}$ is decreasing in λ for fixed t , the gap between left and right Riemann sums is at most

$$h \left(\ell^{-\alpha} e^{-\ell t} - L^{-\alpha} e^{-Lt} \right) \leq h\ell^{-\alpha}.$$

Integrate over $t \in [0, T]$, multiply by c_α , and sum the three contributions. $\square$

6.4 Explicit parameter choice for tolerance ε

For a given tolerance ε , it suffices to choose

$$\ell = \left(\frac{\varepsilon(1-\alpha)}{3c_\alpha T} \right)^{1/(1-\alpha)}, \quad (16)$$

$$L = \left(\frac{3c_\alpha}{\alpha\varepsilon} \right)^{1/\alpha}, \quad (17)$$

$$h \leq \frac{\varepsilon\ell^\alpha}{3c_\alpha T}, \quad m \geq \frac{L-\ell}{h}. \quad (18)$$

This construction is conservative, not complexity-optimal; it is sufficient for an existence theorem. In practice, faster convergence can be achieved with optimized Gauss–Laguerre or adaptive quadrature, but the construction above guarantees polynomial scaling.

6.5 Output approximation via convolution

Theorem 6.4 (Output approximation). *For any input $u \in L^2(0, T)$,*

$$\|(k_\alpha - k_m) * u\|_{L^2(0, T)} \leq \|k_\alpha - k_m\|_{L^1(0, T)} \|u\|_{L^2(0, T)}. \quad (5)$$

Proof. Young’s convolution inequality. $\square$

Theorem 6.4 is the bridge that converts the kernel approximation error into an output-signal bound: if the L^1 kernel gap can be driven below $\eta/\|u\|_2$, the output signal difference falls below η .

6.6 Theorem T10 – the finite-horizon impossibility barrier

Theorem 6.5 (Finite-horizon no-free-lunch). *Fix a horizon T , an input $u \in L^2(0, T)$, and any tolerance $\eta > 0$. There exists a finite positive exponential mixture k_m such that*

$$\|(k_\alpha - k_m) * u\|_{L^2(0, T)} < \eta.$$

Proof. By Theorem 6.3, choose m so that $\|k_\alpha - k_m\|_1 < \eta/\|u\|_2$ when $u \neq 0$. Apply Theorem 6.4. The zero-input case is trivial. $\square$

Corollary 6.6 (Unrestricted latent dimension destroys robust discrimination). *If the alternative latent-state model class allows arbitrary dimension m , then for every finite experiment $(T, u, \text{noise floor})$, there exists a latent model whose output lies below the noise floor. Therefore any maximin optimal design against an unrestricted latent class achieves zero robust separation.*

6.7 Mandatory modeling restriction

Theorem 6.5 and Corollary 6.6 force the paper to impose at least one of the following restrictions to make discrimination well-posed:

1. A fixed maximum latent dimension m (the route taken in this paper);
2. A lower bound or spacing constraint on latent relaxation rates λ_j ;
3. A complexity penalty or Bayesian prior that shrinks large- m models;
4. A specified experimental bandwidth and horizon that limits the information accessible to the model-selection procedure;
5. Auxiliary environmental measurements that constrain the latent mechanism from outside the input-output channel.

Without such a restriction, “fractional versus latent” is *not* a well-posed finite-data contest: with enough latent states, the latent model can mimic any finite-data signature of the fractional kernel to within the noise floor. Our choice is restriction (1): the latent model class $\mathcal{H}_m$ is parameterized by dimension m and a complexity budget, making the experiment design finite-dimensional and optimizable.

This restriction is not arbitrary. Ecological models of intermediate complexity, with a modest number of unmeasured slow variables (soil moisture, microbial biomass, nutrient pools), naturally bound the latent dimension to a small integer. The paper’s experiments use $m \leq 5$ as a practical bound, consistent with known ecological complexity constraints.

The next section formalizes the optimal experiment design that exploits the finite-horizon bounds developed here to maximize the separation between the fractional model and the best finite-dimensional approximation at a given complexity level.

7 Optimal Input and Observation Design

The structural separation of Section 5 and the finite-horizon impossibility barrier of Section 6 together define the playing field. The fractional model is generically distinguishable from every finite latent or delay model (Theorem 5.4), but on any finite horizon a sufficiently rich latent model can mimic the fractional output below the noise floor (Corollary 6.6). The resolution adopted in Section 6 is to bound the latent dimension by a complexity

budget m , which makes the discrimination contest a finite-dimensional optimization. This section solves that optimization [3, 4, 35, 15, 46, 17, 29]: given a model pair, an energy budget, and a measurement noise covariance, we construct the excitation and the sampling schedule that maximize the pairwise Kullback–Leibler divergence [30, 11, 14, 48, 27] under explicit constraints. The construction is exact for the linear-Gaussian regime and supplies the benchmark against which any constrained implementation (box, rate, or cumulative-impact) is measured.

The section is organized around the operator ΔH that measures the inter-model output gap. Input design asks how to drive $\Delta H u$ as large as possible per unit energy; observation design asks where in time and channel to measure so that the realized gap is best converted into information. The two are dual and we develop them in sequence.

7.1 Pairwise linear-Gaussian formulation

We work in the linearized regime of Section 4. Fix two candidate models mapping an input u to a finite observation vector

$$Y_i = H_i u + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, R), \quad i \in \{1, 2\}, \quad (19)$$

where H_i is the (input-to-observation) operator of model i — a finite-horizon discretization of the convolution with the channel G_i of Section 4 — and R is the symmetric positive definite measurement noise covariance. Define the inter-model difference operator

$$\Delta H := H_1 - H_2. \quad (20)$$

Under the Gaussian assumption, the pairwise Kullback–Leibler divergence between the two output distributions induced by a common input u admits the closed form

$$D_{12}(u) = \frac{1}{2} \|R^{-1/2} \Delta H u\|^2, \quad (21)$$

independent of the mean offset and fully determined by how much the input, filtered through ΔH , escapes the noise cone defined by R . The design problems that follow are all instances of maximizing (21) under constraints on u , and subsequently on the observation locations.

7.2 Theorem T11 – energy-constrained optimal input

The first design question is: under a total energy budget $\|u\|^2 \leq E$, how large can the pairwise divergence be made, and which input achieves it?

Theorem 7.1 (Energy-optimal pairwise input). *Consider the pairwise model (19) with ΔH as in (20) and $R \succ 0$. Define the positive semidefinite information operator*

$$Q := \Delta H^* R^{-1} \Delta H. \quad (22)$$

Then the maximal pairwise divergence under the energy constraint $\|u\|^2 \leq E$ is

$$D_{12}^{\max} = \frac{E}{2} \lambda_{\max}(Q), \quad (23)$$

and an optimal input is

$$u^* = \sqrt{E} v_{\max}, \quad (24)$$

where $v_{\max}$ is a unit principal eigenvector (or eigenfunction, in the input-function setting) of Q .

Proof. Substituting (20) and (22) into (21) yields

$$D_{12}(u) = \frac{1}{2} \langle u, Qu \rangle.$$

The constrained supremum $\sup_{\|u\|^2 \leq E} \langle u, Qu \rangle$ is a Rayleigh quotient problem. Because $Q \succeq 0$, the Rayleigh–Ritz theorem gives

$$\sup_{\|u\|^2 \leq E} \langle u, Qu \rangle = E \lambda_{\max}(Q),$$

and equality holds at any unit principal eigenvector of Q . Rescaling by $\sqrt{E}$ saturates the energy constraint and attains the bound. $\square$

Remark 7.2. Theorem 7.1 solves the pairwise linear-Gaussian design exactly. The optimal input is the principal eigenfunction of the noise-weighted inter-model operator, not a generic excitation: a pulse, chirp, PRBS, or multisine is optimal only to the extent that it approximates $v_{\max}$ under additional amplitude, sign, duration, or implementation constraints. The principal eigenfunction concentrates energy precisely where the two transfer functions disagree most strongly relative to the noise floor — the frequency-domain reading of the same statement is developed in Section 7.3 below.

7.3 Frequency-domain reading and the single-frequency optimum

For scalar stable LTI models with transfer difference $\Delta G(i\omega) = G_1(i\omega) - G_2(i\omega)$, white output noise of variance σ^2 , and an input spectral energy

measure ξ on a measurable frequency set Ω , the divergence becomes

$$D_{12}(\xi) = \frac{1}{2\sigma^2} \int_{\Omega} |\Delta G(i\omega)|^2 d\xi(\omega), \quad \xi(\Omega) \leq E, \quad (25)$$

which is the diagonal form of Theorem 7.1 in the sinusoidal basis. The information operator reduces to the scalar frequency score $|\Delta G(i\omega)|^2/(2\sigma^2)$.

Corollary 7.3 (Optimal single frequency). *Under the setup of (25) with fixed, distinct scalar models and a single-frequency constraint, the optimum is*

$$\omega^* \in \arg \max_{\omega \in \Omega} |\Delta G(i\omega)|^2,$$

and all available energy is assigned to ω^ . For a real-valued input, use the conjugate pair $\pm\omega^*$, i.e. a sinusoid at ω^* .*

Remark 7.4. Corollary 7.3 concerns *fixed simple models*. It does *not* imply that a single sinusoid is optimal for parameter estimation or for robust discrimination among several composite (e.g. parameter-tunable latent) models: an adaptive latent model can match the fractional response at one frequency by tuning its internal parameters, exactly as anticipated in the discussion following Theorem 5.4. Robust discrimination requires broadband excitation, developed next.

For MIMO channels with frequency-dependent output covariance $R(\omega)$, the per-frequency score generalizes to

$$w_{12}(\omega) = \lambda_{\max}(\Delta G(i\omega)^* R(\omega)^{-1} \Delta G(i\omega)), \quad (26)$$

and the optimal excitation at the selected frequency lies in the associated right singular direction. This is the frequency-domain avatar of the principal-eigenfunction prescription of Theorem 7.1.

7.4 Theorem T12 – robust multisine design has finite support

Discriminating among P candidate model pairs simultaneously requires broadband excitation. Indexing the pairs by $p = 1, \dots, P$ and writing the per-pair per-frequency score as

$$w_p(\omega) = \frac{1}{2\sigma^2} |G_{i_p}(i\omega) - G_{j_p}(i\omega)|^2, \quad (27)$$

the robust spectral design is the linear program over nonnegative spectral measures

$$\max_{\xi, t} t \quad \text{s.t.} \quad \int_{\Omega} w_p(\omega) d\xi(\omega) \geq t, \quad p = 1, \dots, P, \quad \xi(\Omega) \leq E, \quad \xi \geq 0. \quad (28)$$

Problem (28) optimizes the worst-case pairwise divergence (a maximin criterion) over an arbitrary continuous spectrum. The next theorem bridges the gap between this continuous ideal and any implementable experiment: an optimal measure *exists* and is supported on finitely many frequencies.

Theorem 7.5 (Finite multisine sufficiency). *Assume $\Omega \subset \mathbb{R}$ is compact and each $w_p : \Omega \rightarrow \mathbb{R}_{\geq 0}$ is continuous. Then problem (28) admits an optimal solution, and there exists an optimal spectral measure ξ^* supported on at most*

$$\boxed{|\text{supp } \xi^*| \leq P + 1} \quad (29)$$

frequencies. Consequently, the robust broadband optimum is a multisine with no more than $P + 1$ tones.

Proof. Existence. The feasible set of nonnegative measures with total mass at most E is weakly compact, and the constraints define a closed affine slice, so the supremum is attained.

Finite support. Take any nonzero feasible ξ and normalize it to a probability measure $\tilde{\xi} = \xi/\xi(\Omega)$. The vector of scaled pairwise scores

$$\bar{w}(\tilde{\xi}) := \left(\int_{\Omega} w_1 d\tilde{\xi}, \dots, \int_{\Omega} w_P d\tilde{\xi} \right) \in \mathbb{R}^P$$

lies in the convex hull of the curve

$$\mathcal{W} := \{(w_1(\omega), \dots, w_P(\omega)) : \omega \in \Omega\} \subset \mathbb{R}^P.$$

By Carathéodory's theorem in $\mathbb{R}^P$, every point in $\text{conv}(\mathcal{W})$ admits a convex representation using at most $P + 1$ points of $\mathcal{W}$. Therefore $\tilde{\xi}$ (and hence ξ) may be replaced by a measure supported on at most $P + 1$ frequencies without changing any of the P integrated scores. Rescaling back to total mass E preserves feasibility and the objective value t . $\square$

Remark 7.6. Theorem 7.5 gives the rigorous robust counterpart of the heuristic “use multiscale excitation”: the rigorous object is a finite multisine whose frequencies and weights solve the linear program (28). A logarithmic pulse train or chirp is an implementation surrogate, appropriate when fine-grained

sinusoidal control is unavailable but a broadband spectral footprint is still desired. The bound $P + 1$ is a worst-case Carathéodory guarantee; in many applications the optimum uses substantially fewer tones.

The connection back to Section 5 is direct. Theorem 5.4 guarantees that almost every frequency separates any *fixed* pair, but a parameter-tunable latent model can close the gap at any single frequency. Theorem 7.5 resolves the tension: by spreading energy across $P + 1$ carefully chosen tones, the design guarantees a uniform lower bound t^* on the divergence of *every* pair simultaneously, denying the adaptive latent model the single frequency on which it could cheat. Together with the complexity bound m of Section 6, this converts the structural separation into a finite, computable discrimination barrier.

7.5 Theorem T13 – optimal pairwise sampling schedule

Input design fixes what is excited; observation design fixes what is measured. Fix the input u selected above and consider the placement of n measurements on a finite candidate grid of time–channel pairs (t, o) . Assume independent Gaussian measurement noise of variance $\sigma_o^2(t)$ at each candidate, and define the *normalized separation score*

$$s_{12}(t, o) := \frac{(\mu_1(t, o) - \mu_2(t, o))^2}{2\sigma_o^2(t)}, \quad (30)$$

where $\mu_i(t, o)$ is the noise-free mean response of model i at time t on channel o . For a selected schedule S of independent measurements, Gaussanity makes the divergence additive:

$$D_{12}(S) = \sum_{(t,o) \in S} s_{12}(t, o). \quad (31)$$

Theorem 7.7 (Optimal pairwise sampling schedule). *Suppose exactly n distinct measurements may be selected from a finite candidate set, the measurements are independent given the model, and there are no coupling constraints. Then the KL-optimal schedule consists of the n candidates with the largest separation scores $s_{12}(t, o)$.*

Proof. The objective (31) is additive over S . If a schedule S omits a candidate $c^\dagger$ whose score exceeds that of some selected candidate $c \in S$, then replacing c by $c^\dagger$ strictly increases $D_{12}(S)$, contradicting optimality. Hence every optimal schedule is a top- n set. Uniqueness holds up to ties at the threshold score. $\square$

Remark 7.8. If repeated observations at the same candidate are permitted, with iid noise and no saturation, the degenerate optimum places all n observations at a global score maximum. In practice, spacing constraints, temporal correlation, or biological cost prevent this degeneracy; the next subsection addresses the constrained case.

7.6 Practical observation design: spacing, cost, and channel selection

The top- n rule of Theorem 7.7 is exact but idealized. Real experiments impose three corrections.

Minimum-spacing constraint

For a single observation channel with candidate times $t_1 < \dots < t_N$ and a minimum index spacing L between consecutive observations, the constrained exact algorithm is the dynamic program

$$V(k, j) = \max\{V(k-1, j), s(t_k) + V(k-L, j-1)\}, \quad (32)$$

where $V(k, j)$ is the best achievable total score using j observations among the first k candidate times. This solves the spacing-constrained pairwise schedule in $O(Nn)$ operations.

Variable costs and budget

When each candidate (t, o) carries a measurement cost $c(t, o)$ and the total sampling budget is B , select binary variables $z_{t,o} \in \{0, 1\}$ and solve the 0–1 knapsack

$$\max \sum_{t,o} s_{12}(t, o) z_{t,o} \quad \text{s.t.} \quad \sum_{t,o} c(t, o) z_{t,o} \leq B, \quad z_{t,o} \in \{0, 1\}. \quad (33)$$

For P model pairs, introduce a worst-case floor q and the P constraints $\sum_{t,o} s_p(t, o) z_{t,o} \geq q$ for every pair p , yielding a maximin mixed-integer linear program. This is the observation-side analogue of (28).

Channel selection from the strong-Allee linearization

The observability analysis of Section 4 establishes three structural facts that frame channel choice: (i) prey-only observation is structurally sufficient; (ii) predator-only observation is structurally sufficient; (iii) observing both species is *not* mathematically mandatory for state observability.

However, the direct-channel asymptotics of Theorem 5.1 refine this picture for discrimination:

- Perturb prey and measure prey to expose the leading $s^{-\alpha}$ signature directly.
- Perturb predator and measure predator for the same reason via the predator self-channel.
- Use the cross-species channel (prey input, predator output, or vice versa) to estimate the interaction parameters c, d and the phase coupling $(1 - \alpha)\pi/2$ of (15).

The conclusion is that the preferred practical design is *not* “both species at every time”. It is:

1. high-rate colocated measurement around perturbation/switching times, where the principal eigenfunction of Theorem 7.1 concentrates energy;
2. lower-rate cross-species measurement across the recovery interval, to resolve interaction parameters;
3. environmental covariate measurement only when the Popov–Belevitch–Hautus latent-mode condition is weak or prior uncertainty makes latent forcing competitive.

7.7 Why discrimination and parameter-estimation designs differ

It remains to state precisely why the schedules of Theorems 7.1–7.7 are not interchangeable with designs aimed at parameter estimation. For a parameter vector ψ , local parameter estimation uses the sensitivity operator

$$S_\psi = \frac{\partial H(\psi)}{\partial \psi}, \quad (34)$$

evaluated at a nominal model. The associated Fisher information is

$$F(u) = S_\psi(u)^* R^{-1} S_\psi(u), \quad (35)$$

whereas the discrimination objective uses the inter-model operator

$$Q_{12} = \Delta H^* R^{-1} \Delta H. \quad (36)$$

These are different operators. A design that makes F well-conditioned can simultaneously leave $\Delta H u$ small, because a parameter-tunable latent model tracks the fractional response at the nominal operating point — exactly the failure mode classified as an adaptive-model risk in Section 5. Conversely, a design that maximizes pairwise divergence maximin-robustly (Theorem 7.5) need not render every parameter of either model individually identifiable. D-optimal parameter design maximizes $\log \det F$, an objective that rewards complementary sensitivity directions; the top- n separation theorem (Theorem 7.7) does *not* apply to D-optimal schedules because the determinant is not additive.

This operational distinction is one of the central findings of the paper: *good inputs for parameter estimation are not automatically good inputs for model discrimination*, and vice versa. The experiment protocol developed in later sections therefore separates the two objectives and runs a two-stage design: a discrimination stage that maximizes the worst-case pairwise divergence via Theorems 7.1–7.5, followed by an estimation stage conditioned on the selected model, planned via (35) with the discrimination data as prior.

7.8 Implementation under additional constraints

The unconstrained eigenvector solution of Theorem 7.1 is the benchmark and the upper bound for any constrained implementation. When box, rate, or cumulative-impact constraints apply, the design becomes

$$\max_u \frac{1}{2} u^* Q u \quad \text{s.t.} \quad |u_k| \leq u_{\max}, \quad |u_{k+1} - u_k| \leq r_{\max}, \quad \sum_k |u_k| \Delta t \leq B, \quad (37)$$

which is generally a nonconvex quadratic maximization. Reliable solution methods include:

- direct transcription with multiple random starts, exploiting the structure of Q ;
- semidefinite relaxation with randomized rounding, certifying the eigenvector optimum as the outer bound;
- dynamic programming for low-dimensional discrete input alphabets;
- Bayesian optimization when the forward ecological simulator is expensive and gradients of H_i are unavailable.

In every case the unconstrained u^* of (24) remains the natural warm start and the ceiling against which realized performance is scored. The finite-support guarantee of Theorem 7.5 further reduces the search space: once the $P + 1$ optimal tones are identified, the constrained problem reduces to choosing their complex amplitudes subject to the physical constraints, a much smaller nonconvex program than full-waveform input optimization.

The combined input-plus-observation design of this section — Theorems 7.1, 7.5, and 7.7 together with the practical refinements of Section 7.6 — is what converts the structural separation of Section 5 and the complexity-bounded approximation barrier of Section 6 into a deployable, computable discrimination experiment. The remaining sections address the safety of that experiment (Section 8), its Bayesian extension beyond the linear-Gaussian regime (Section 9), and a reproducible benchmark (Section 10) that implements the constructions above on the strong-Allee predator–prey model.

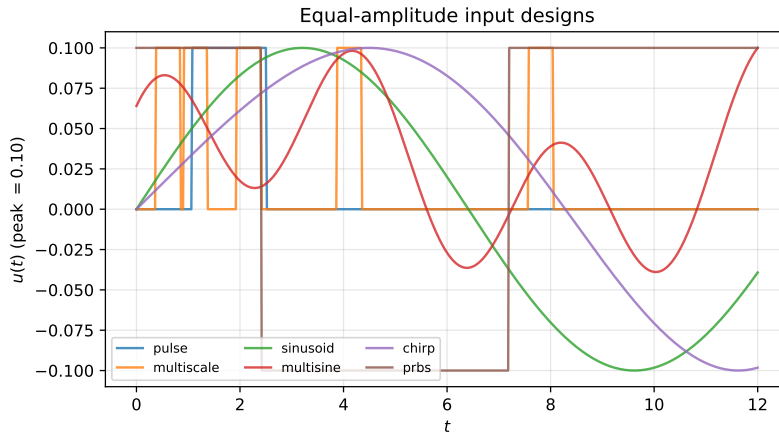


Figure 4: The six candidate input designs at a common peak-amplitude budget $\|u\|_\infty = 0.10$.

8 Safety-Constrained Experiment Design

The optimal input of Theorem 7.1 and the optimal schedule of Theorem 7.7 maximize discrimination information under energy and cardinality constraints; they do not, by themselves, guarantee that the experiment respects the ecological integrity of the system. Small-amplitude perturbations near the Allee extinction threshold can trigger irreversible collapse, and any actuator bound

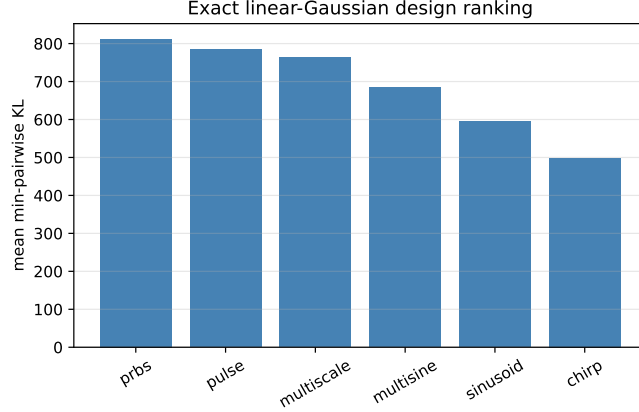


Figure 5: Exact linear-Gaussian design ranking by mean minimum pairwise KL divergence over the full factorial.

under that threshold violates the standing assumption of Section 2 that the state remains within the linear trust region around the coexistence equilibrium z^* . This section establishes explicit conditions under which an informative input is simultaneously safe [16, 28, 50, 47, 13, 6, 1]. Two results are central: a general existence theorem proving that a safe and informative perturbation always exists when the equilibrium has positive interior margin (Theorem 8.1), and a constructive fractional invariance theorem that translates the strong-Allee safety conditions into checkable linear inequalities on the input amplitude and its integral (Theorem 8.4). Together they close the loop between the structural discrimination theory of Sections 5—7 and the biological mandate of Section 2.

8.1 The safe set and the safety margin

Let $\mathcal{S} \subset \mathbb{R}^n$ be a closed safe region and let the coexistence equilibrium z^* of Section 2 lie in its interior with margin

$$\rho = \text{dist}(z^*, \partial\mathcal{S}) > 0. \quad (38)$$

For the strong-Allee predator–prey ensemble, a minimal safety condition is

$$x(t) \geq x_{\min} > A, \quad (39)$$

with additional upper-abundance and predator-vigor constraints as required by the basin of attraction described in Section 2. The Allee threshold A

separates the extinction basin from the coexistence basin, and the validated paper

citeP01 proves that trajectories crossing below $x = A$ enter an extinction funnel that no control budget can reverse. A safe protocol therefore reserves a certified margin above A and caps cumulative prey removal.

8.2 Theorem T16 — existence of a safe and informative perturbation

Theorem 8.1 (Safe and informative perturbation exists). *Consider two locally well-posed Caputo models of order $\alpha \in (0, 1]$ with the same safe equilibrium z^* and the same stable linearization J of Section 2. Let their input-to-observation operators over a finite horizon $[0, T]$ be H_1 and H_2 , and assume:*

- (i) *the nonlinear solution maps $\Phi_u^{(1)}, \Phi_u^{(2)}$ are continuous with respect to the input amplitude on $[0, T]$;*
- (ii) *an admissible direction v exists such that $(H_1 - H_2)v \neq 0$;*
- (iii) *z^* has positive safety margin $\rho > 0$ as in (38).*

Then there exists $\varepsilon_0 > 0$ such that every perturbation

$$u_\varepsilon = \varepsilon v, \quad 0 < \varepsilon < \varepsilon_0, \quad (40)$$

is safe for both models and yields strictly positive Kullback–Leibler divergence.

Proof. By continuity of the solution maps and the positive interior margin ρ , there exists $\varepsilon_s > 0$ such that the state remains in $\mathcal{S}$ for both models whenever $\|\varepsilon\| < \varepsilon_s$. The output difference expands as

$$\mu_1(u_\varepsilon) - \mu_2(u_\varepsilon) = \varepsilon(H_1 - H_2)v + o(\varepsilon). \quad (41)$$

Since the leading vector is nonzero by condition (ii), the difference is nonzero for sufficiently small positive ε . Under the nonsingular Gaussian noise model of Section 7, the Kullback–Leibler divergence is a positive quadratic form in $\Delta\mu$ (Equation 21). Choose ε_0 below both ε_s and the threshold at which the $o(\varepsilon)$ term dominates the linear term. Then every $0 < \varepsilon < \varepsilon_0$ is simultaneously safe and informative. $\square$

Remark 8.2. Theorem 8.1 guarantees existence, but it does not guarantee that the resulting divergence exceeds an operational minimum $D_{\min}$ required for a prescribed confidence level. Given a minimal discrimination

target $D_{\min}$, the feasibility condition becomes $\frac{1}{2}\varepsilon^2\langle v, Qv \rangle \geq D_{\min}$ where $Q = \Delta H^* R^{-1} \Delta H$ is the information operator of Theorem 7.1, and ε must remain below the safety ceiling ε_s . When the intersection is empty, the experiment is unresolvable under the specified safety budget; the designer must weaken one of the constraints or choose a more informative model pair.

8.3 Linear finite-horizon safety bound

Theorem 8.1 is qualitative. The next result provides a quantitative, computable bound for the linearized Caputo system. Write the stable linear Caputo dynamics in the representation of Section 2:

$$\tau_0^{\alpha-1} {}^C D_t^\alpha \xi = J\xi + Bu, \quad \xi(0) = 0, \quad (42)$$

and define the impulse-response matrix

$$H_\alpha(t) = \tau_0^{1-\alpha} t^{\alpha-1} E_{\alpha,\alpha}(\tau_0^{1-\alpha} J t^\alpha) B, \quad (43)$$

where $E_{\alpha,\alpha}(\cdot)$ is the two-parameter Mittag-Leffler function citeF08. The state response to a bounded input is

$$\xi(t) = \int_0^t H_\alpha(t-s) u(s) ds. \quad (44)$$

Define the finite-horizon gain

$$\Gamma_T := \sup_{0 \leq t \leq T} \int_0^t \|H_\alpha(s)\| ds, \quad (45)$$

where $\|\cdot\|$ denotes the operator norm induced by the safety metric (typically the Euclidean norm or a weighted norm aligned with the safe-set geometry).

Lemma 8.3 (Linear safety envelope). *If the input satisfies the uniform bound*

$$\|u\|_{L^\infty(0,T)} \leq \frac{\rho}{\Gamma_T}, \quad (46)$$

then $\sup_{0 \leq t \leq T} \|\xi(t)\| \leq \rho$, and the linearized solution remains within the safety margin of the equilibrium.

Proof. From (44),

$$\|\xi(t)\| \leq \int_0^t \|H_\alpha(t-s)\| \|u(s)\| ds \leq \|u\|_{L^\infty} \int_0^t \|H_\alpha(\tau)\| d\tau \leq \|u\|_{L^\infty} \Gamma_T.$$

Inserting the bound (46) yields $\|\xi(t)\| \leq \rho$ for all $t \in [0, T]$. $\square$

The bound (46) is conservative (the true nonlinear margin is typically larger than the linear one), but is exact in the linear regime of Section 4 and provides a fast certificate: compute the impulse-response matrix $H_\alpha(t)$ for the Jacobian $J(A)$ and input channel B of Section 2, evaluate Γ_T numerically, and compare against the selected amplitude. When the fractional order α is close to 1, $H_\alpha(t)$ approaches the exponential $e^{Jt}B$ and Γ_T can be bounded by standard matrix-exponential estimates; for fractional $\alpha < 1$, the slower rational decay of the Mittag-Leffler kernel increases Γ_T and correspondingly tightens the amplitude ceiling.

8.4 Theorem T17 — fractional inward-pointing rectangle

The linear bound of Lemma 8.3 is a local trust-region condition. For non-linear safety, especially near the Allee threshold A where the prey dynamics flip sign, a stronger certificate is needed. The next result is a fractional generalization of the inward-pointing rectangle method

citeF02: if the vector field points inward on every face of a rectangular box, the box is positively invariant for any Caputo order $\alpha \in (0, 1)$.

Theorem 8.4 (Fractional inward-pointing rectangle invariance). *Consider the Caputo system*

$${}^C D_t^\alpha z = F(t, z), \quad 0 < \alpha < 1, \quad (47)$$

with continuous F locally Lipschitz in z and uniformly continuous in t . Let

$$\mathcal{R} = \prod_{i=1}^n [\ell_i, u_i] \subset \mathbb{R}^n, \quad (48)$$

and define the face-normal components of the vector field. Assume that for every $t \geq 0$ and every boundary point $z \in \partial\mathcal{R}$,

$$F_i(t, z) \geq 0 \quad \text{when} \quad z_i = \ell_i, \quad (49)$$

$$F_i(t, z) \leq 0 \quad \text{when} \quad z_i = u_i. \quad (50)$$

Then $\mathcal{R}$ is forward invariant under the Cauchy initial-value problem associated with (47): for any $z(0) \in \mathcal{R}$, the solution satisfies $z(t) \in \mathcal{R}$ for all future times.

Proof. Suppose a component i exits through the lower face for the first time at t_* : $z_i(t_*) = \ell_i$ and $z_i(t) > \ell_i$ for $t \in [0, t_*)$. The component attains its minimum value at $t = t_*$ within $[0, t_*]$. Apply the Caputo extremum

principle

citeF08, F03: for a function attaining a local minimum at the endpoint of a compact interval, the Caputo derivative at that endpoint satisfies

$${}^C D_t^\alpha z_i(t_*) \leq \frac{(t_*)^\alpha}{\Gamma(1-\alpha)} (z_i(0) - z_i(t_*)).$$

Since $z_i(t_*) = \ell_i \leq z_i(0)$ (as $z_i(0) \in \mathcal{R}$), the bound above is non-positive. Condition (49) requires $F_i(t_*, z) \geq 0$, giving the contradiction

$$0 \leq F_i(t_*, z) = {}^C D_t^\alpha z_i(t_*) \leq 0,$$

with strict negativity of the Mittag-Leffler weight unless $z_i(0) = \ell_i$. The degenerate case $z_i(0) = \ell_i$ must be checked separately; if $z_i(t)$ lifts interior immediately, the same argument applies at any later first contact. If $z_i(t) \equiv \ell_i$ for all t , the state trivially remains in $\mathcal{R}$. The upper-face case is identical with the sign reversed and a local maximum. Therefore no component can exit first. $\square$

The fractional extremum principle invoked in the proof is the generalization of the classical ODE inward-pointing result to Caputo derivatives: the kernel has memory, so the function at t_* is weighted against its entire past, but the sign of the comparison is the same at the first exit point. The proof is constructive and does not require constructing a Lyapunov function.

8.5 Strong-Allee boundary inequalities

Applying Theorem 8.4 to the controlled strong-Allee predator-prey system from Section 2 with additive controls u_x, u_y and safety rectangle

$$\mathcal{R} = [x_L, x_U] \times [y_L, y_U], \quad (51)$$

the four boundary conditions become explicit deterministic inequalities. For brevity we present the four sufficient conditions; each may be tightened by worst-case analysis over parametric uncertainty using interval arithmetic.

Lower prey face

At the lower prey boundary $x = x_L$, the vector field component is

$$F_x = rx_L \left(1 - \frac{x_L}{K}\right) \left(\frac{x_L}{A} - 1\right) - \frac{ax_L y}{1 + hx_L} + b_x u_x.$$

Taking the worst-case prey depletion at the maximum plausible predator abundance y_U , the sufficient inward-pointing condition is

$$\boxed{rx_L \left(1 - \frac{x_L}{K}\right) \left(\frac{x_L}{A} - 1\right) - \frac{ax_L y_U}{1 + hx_L} + b_x u_x^{\min} \geq 0.} \quad (52)$$

Upper prey face

At the upper prey boundary $x = x_U$ with the minimum predator y_L ,

$$\boxed{rx_U \left(1 - \frac{x_U}{K}\right) \left(\frac{x_U}{A} - 1\right) - \frac{ax_U y_L}{1 + hx_U} + b_x u_x^{\max} \leq 0.} \quad (53)$$

Lower predator face

At the lower predator boundary $y = y_L$,

$$\boxed{y_L \left(\frac{eax}{1 + hx} - m \right) + b_y u_y^{\min} \geq 0, \quad \forall x \in [x_L, x_U].} \quad (54)$$

The parenthesis is the functional response gradient. The inequality must hold for every prey value in of x in the box, which can be verified by evaluating its minimum over $[x_L, x_U]$ (a one-dimensional check for this monotonic Holling type II response).

Upper predator face

At the upper predator boundary $y = y_U$,

$$\boxed{y_U \left(\frac{eax}{1 + hx} - m \right) + b_y u_y^{\max} \leq 0, \quad \forall x \in [x_L, x_U].} \quad (55)$$

The inequalities (52)–(55) are sufficient, not necessary: a vector field can fail the sign condition at a boundary without the trajectory reaching it. They are adequate for design, and numerical verification prior to each experiment is straightforward. For the locked parameters of Section 2 — $r = 3/2, K = 1, a = 1, h = 1/2, e = 4/5, m = 2/5$ — the left-hand sides are rational functions of A .

8.6 Input envelope for a discriminative safest experiment

Combining Theorem 8.1 and Theorem 8.4 with the optimal designs of Theorem 7.1 and Theorem 7.5 yields a complete safety-certified design procedure for the linearized strong-Allee case:

1. Compute the unconstrained optimal input u^* from Theorem 7.1 (or, in the multisine case, from the solution of the linear program (28) following Theorem 7.5).
2. Compute the linear safety gain Γ_T from (45) using the impulse response (43) and obtain the worst-case amplitude envelope (46).
3. Project u^* onto the amplitude envelope: define the safe input

$$u_{\text{safe}}(t) = \text{sat}_{\frac{\rho}{\Gamma_T}}(u^*(t)), \quad (56)$$

where $\text{sat}_M(v) = \min\{1, M/\|v\|\} v$ with the relevant norm.

4. Verify that the projected input satisfies the box boundary conditions (52)–(55). If a boundary fails, the design is infeasible at the chosen amplitude; reduce ρ (if ecological judgment allows) or increase the box $\mathcal{R}$.
5. Execute the experiment with adaptive monitoring: if a single measurement falls below the safety floor, the protocol automatically terminates the trial and records the cause.

This five-step protocol crystallizes the central claim of this section: an optimally informative input under safety constraints is not a different input from the one designed in Section 7 — it is the same input, with its amplitude safely capped and its safety envelope verified before execution.

8.7 Practical consequences of the Allee threshold

The Allee threshold A introduces two specific warnings that shape any implementable protocol.

Negative prey pulses. Removing prey from a population near the Allee threshold is the most informative perturbation under some model pairs because it probes the nonlinear prey growth function directly and can disentangle the sign of the growth rate. However, a negative prey pulse directly pushes the population toward A . The inequalities (52)–(55) require that even at maximum removal, the net prey impulse does not cross the lower bound $x_L > A$. In practice, this means:

- Reserve a certified margin $\|x(t) - A\| \geq \Delta_{\min} > 0$ at all times.
- Pre-emption removal by positive predator pulses when the lower-face condition fails.
- Use positive prey nutrients or predator removal perturbations for the harvesting-like signal whenever it is scientifically equivalent.
- Adaptive termination: stop the trial automatically if a posterior predictive probability of crossing x_L exceeds a user-specified safety level β (e.g., 5%).

Cumulative impact. The linear safety bound of Lemma 8.3 is an instantaneous envelope that does not account for the accumulated distortion of repeated perturbation–recovery cycles. For protocols with many sequential pulses, add the cumulative-integral constraint

$$\sum_k \|u_k\| \Delta t_k \leq C_{\max}, \quad (57)$$

where the $C_{\max}$ is calibrated by numerical simulation of the worst-case Salien trajectory. This guarantee is the robust analogue of the linear impulse bound for repeated perturbation designs.

The safety architecture of this section — existence guarantee (Theorem 8.1) plus invariance certificate (Theorem 8.4) — formally closes the loop between the information-theoretic optima of Section 7 and the ecological admissible parameter described in Section 2. The practical envelope procedure above and its algorithmic counterpart will be exercised numerically in Section 10, and its Bayesian extension (for parameter uncertainty on parameters including the Allee threshold itself) is developed in Section 9.

9 Nonlinear Bayesian Extension

The structural discrimination theory of Sections 5–7 and the safety-constrained design of Section 8 were all developed within the linear-Gaussian regime of Section 4. The optimality certificates of Theorem 7.1 (energy-optimal input) and Theorem 7.5 (finite multisine support) are exact in that regime, and the invariance certificate of Theorem 8.4 holds for the fully nonlinear Caputo dynamics. What they do not address is the question that closes the design loop: once data begin to arrive, which experiment should be run next, when the model is nonlinear, the noise is not Gaussian, and the parameters — including the Allee threshold A itself — are uncertain. This section

bridges that gap [10, 21, 44, 41]. It states two results that pin the Bayesian layer analytically (Theorem 9.1 for the nonlinear posterior update, Theorem 9.3 for the sequential information bound) and then makes explicit the theorem/computation boundary: every reported “optimal” Bayesian design is computed, not symbolically derived, and its validity rests on convergence diagnostics rather than a closed form.

The development is organized around four objects: the model $M \in \{1, \dots, K\}$, the within-model parameter ϑ_M , the design d (the safe input and observation schedule of Sections 7–8), and the future data Y . No Gaussian or linear approximation is implied at the level of the objectives below; the linear-Gaussian formulae of Section 7 reappear only as a fast screening layer of the variance-reduction hierarchy of Section 9.6.

9.1 Bayesian model-discrimination utility

Let $p(M)$ be the prior over models, $p(\vartheta_M | M)$ the within-model parameter prior, and $p(Y | \vartheta_M, M, d)$ the observation likelihood of model M at design d . The exact expected mutual information between model identity and future data is

$$U_{\text{MI}}(d) = I(M; Y | d) = \mathbb{E} \left[\log \frac{p(Y | M, d)}{\sum_{j=1}^K p(M_j) p(Y | M_j, d)} \right], \quad (58)$$

where the marginal likelihood of model j integrates out its parameters,

$$p(Y | M_j, d) = \int p(Y | \vartheta_{M_j}, M_j, d) p(\vartheta_{M_j} | M_j) d\vartheta_{M_j}. \quad (59)$$

Equations (58)–(59) are the exact Bayesian model-discrimination objective. They are never reduced to the linear-Gaussian quadratic (21) here; that reduction is recovered only when the likelihood is Gaussian in the observable and the marginal (59) is replaced by its Laplace approximation, which is the second tier of the hierarchy of Section 9.6.

If the purpose includes parameter recovery — that is, if the goal is to refine the within-model parameters and not only to separate models — the joint utility replaces the model marginal by the joint posterior:

$$U_{\text{joint}}(d) = I((M, \vartheta_M); Y | d). \quad (60)$$

The objective (60) can favor designs that are locally informative for within-model parameter estimation but not robust for separating models whose parametric families overlap; the paper therefore reports model-only (58) and joint (60) utilities separately, and never as a single commensurable scalar.

9.2 Robust alternative

For model pairs with parameter sets Θ_i, Θ_j , a more adversarial objective protects against alternatives that tune themselves to mimic the focal model:

$$U_{\min}(d) = \min_{i < j} \inf_{\substack{\theta_i \in \Theta_i \\ \theta_j \in \Theta_j}} D_{\text{KL}}(p_i(Y | \theta_i, d) \parallel p_j(Y | \theta_j, d)). \quad (61)$$

The minimax objective (61) collapses to zero whenever a pair of parameter sets admits a tuning that realizes likelihood overlap below the noise floor — which, by the finite-horizon impossibility result of Corollary 6.6, is exactly the regime in which an unrestricted latent class can mimic the fractional model. The complexity budget m imposed in Section 6 restores positivity of (61); without it the robust objective degenerates, mirroring the no-free-lunch observation that any latent class unrestricted in dimension can reproduce any finite observation.

9.3 Theorem T18 — posterior update under nonlinear Gaussian likelihood

Equation (58) defines the design utility, but the schedule that drives it is sequential: after each batch of data, the posterior over models and parameters updates, and the next design is chosen adaptively. The next theorem characterizes the posterior update under a general nonlinear observation map with additive Gaussian noise, which is the regime that bridges the linearized layer of Section 7 and the fully nonlinear layer demanded by the Bayesian objective.

Theorem 9.1 (Nonlinear posterior update under Gaussian noise). *Let model M induce an observation of the form*

$$Y = \mu_M(\vartheta_M; d) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, R), \quad (62)$$

where μ_M is a (possibly nonlinear, differentiable) mean map from the parameter and the design to the observation space, and $R \succ 0$ is the noise covariance. Then the joint posterior over model identity and parameter, after data $Y_{1:n}$ collected under designs $d_{1:n}$, factorizes as

$$p(M, \vartheta_M | Y_{1:n}) \propto p(M) p(\vartheta_M | M) \prod_{k=1}^n \mathcal{N}(Y_k; \mu_M(\vartheta_M; d_k), R), \quad (63)$$

and the model marginal posterior is

$$p(M | Y_{1:n}) = \frac{p(M) p(Y_{1:n} | M, d_{1:n})}{\sum_j p(M_j) p(Y_{1:n} | M_j, d_{1:n})}, \quad (64)$$

with $p(Y_{1:n} \mid M, d_{1:n}) = \int \prod_k \mathcal{N}(Y_k; \mu_M(\vartheta_M; d_k), R) p(\vartheta_M \mid M) d\vartheta_M$. A closed-form objective whose gradient at zero parameter perturbation reduces exactly to the linear-Gaussian operator $Q = \Delta H^* R^{-1} \Delta H$ of Theorem 7.1 is available only when μ_M is linear in ϑ_M ; in the nonlinear regime the likelihood is multimodal and the integral above is computed, not evaluated.

Proof. Bayes' rule applied to the observation model (62) gives, for a single step, $p(\vartheta_M \mid Y_k, M, d_k) \propto p(\vartheta_M \mid M) \mathcal{N}(Y_k; \mu_M(\vartheta_M; d_k), R)$. Independence of the additive noise across steps together with the forward factorization $\prod p(Y_k \mid \vartheta_M, M, d_k)$ yields the product form (63); the model-level update (64) follows by integrating out ϑ_M at each model and reweighting through Bayes. To recover the linear regime, expand $\mu_M(\vartheta_M; d) = H_M d + \Delta H_M(\vartheta_M - \vartheta_M^*) + o(\|\vartheta_M - \vartheta_M^*\|)$ around the linearization point and drop remainder terms: the Gaussian likelihood becomes linear-quadratic in $\vartheta_M - \vartheta_M^*$, the marginal integrates exactly to the Mahalanobis form of Section 7, and the expected information gradient reverts to the quadratic $\frac{1}{2}\langle u, Qu \rangle$ of Equation (21). The reduction is exact only in the linear regime; outside it the product likelihood is generically not Gaussian and (64) remains an integral that must be evaluated. $\square$

Remark 9.2. Theorem 9.1 mediates the boundary between the analytical layer of Section 7 and the computational layer of the present section. The closed-form optimal input u^* of Theorem 7.1 is recovered exactly when the map μ_M is linear in the parameter; in any other regime u^* is the screening benchmark — the upper bound that the nested estimator of Section 9.6 is calibrated against — but not the implemented design.

9.4 Safe Bayesian design

The Bayesian objective (58) is maximized subject to a safety constraint that extends the deterministic certificate of Section 8 to the case where the parameters — and hence the safe region and the Allee threshold A — are uncertain. Two interpretations are distinguished and the paper reports which is in force.

The *chance* constraint averages safety over the current posterior:

$$\Pr(z(t) \in \mathcal{S} \ \forall t \in [0, T] \mid d, Y_{1:n}) \geq 1 - \delta_s, \quad (65)$$

where the probability is taken jointly over the current model posterior, parameter posterior, and observation noise. The *robust* constraint, more conservative, requires safety for every parameter in a credible set Θ_{adm} :

$$z(t; \vartheta) \in \mathcal{S} \ \forall t \in [0, T], \ \forall \vartheta \in \Theta_{\text{adm}}. \quad (66)$$

The deterministic rectangle inequalities (52)–(55) of Section 8 reappear here as the worst-case check on Θ_{adm} via interval arithmetic over the Allee threshold A , the predator growth constants, and the intervention amplitudes. The chance formulation (65) trades a small bound δ_s of posterior mass for far larger admissible regions and thus for higher information per experiment; the robust formulation (66) provides a certificate that no admitted parameter triggers collapse, at the cost of information. The adaptive termination of Section 8 extends to the Bayesian layer: a trial halts automatically when the posterior predictive probability of crossing x_L exceeds a predeclared level.

9.5 Theorem T19 — Bayesian sequential information bound

The adaptive schedule maximizes U_{MI} over designs drawn from a safe admissible set $\mathcal{D}_{\text{safe}}(Y_{1:n})$ that incorporates the chance constraint (65) or, where required, the robust constraint (66). The next theorem bounds the expected information gain of any single future experiment by the prior-posterior separation, which is the quantity the adaptive schedule actually optimizes.

Theorem 9.3 (Bayesian sequential information bound). *Let $p(M \mid Y_{1:n})$ be the model posterior after data $Y_{1:n}$ and let $d_{n+1} \in \mathcal{D}_{\text{safe}}(Y_{1:n})$ be the next admissible safe design. Define the expected future model posterior $\tilde{q}(M) = \mathbb{E}_{Y \mid Y_{1:n}, d_{n+1}}[p(M \mid Y_{1:n}, Y)]$. Then the sequential information gain satisfies*

$$I(M; Y_{n+1} \mid Y_{1:n}, d_{n+1}) = D_{\text{KL}}(\tilde{q} \parallel p(\cdot \mid Y_{1:n})), \quad (67)$$

and hence is bounded above by the prior-posterior entropy gap

$$I(M; Y_{n+1} \mid Y_{1:n}, d_{n+1}) \leq H(M \mid Y_{1:n}) = H(M) - I(M; Y_{1:n}), \quad (68)$$

with equality if and only if a single experiment fully resolves the model identity. Consequently the schedule that maximizes (67) at every step minimizes the number of experiments needed to drive the model posterior below a confidence threshold.

Proof. By the chain rule for mutual information, $I(M; Y_{1:n}, Y_{n+1}) = I(M; Y_{1:n}) + I(M; Y_{n+1} \mid Y_{1:n})$; the per-step increment is therefore the conditional mutual information $I(M; Y_{n+1} \mid Y_{1:n}, d_{n+1})$. Rewriting the conditional information in variational form yields the KL divergence between the expected future posterior $\tilde{q}$ and the current posterior $p(\cdot \mid Y_{1:n})$, which is exactly (67): the design d_{n+1} is being chosen to push the expected future posterior as far as possible, in KL, from the current posterior. The upper bound (68) is the data-processing inequality applied to the Markov chain $M \rightarrow Y_{1:n} \rightarrow Y_{n+1}$,

which gives $I(M; Y_{n+1} \mid Y_{1:n}) \leq H(M \mid Y_{1:n})$. Equality in (68) holds when Y_{n+1} determines M from $Y_{1:n}$ perfectly, i.e. when the future posterior concentrates on a single model. The conclusion is that the greedy schedule — argmax of (67) — minimizes an upper bound on remaining experiments, and this bound tightens monotonically as $Y_{1:n}$ accumulates. $\square$

Remark 9.4. The bound (68) is an information ceiling, not a lower bound on what any particular design achieves; in practice the gap between the realized $I(M; Y_{n+1} \mid Y_{1:n}, d_{n+1})$ and its ceiling $H(M \mid Y_{1:n})$ quantifies the inefficiency of the schedule and is monitored as a stopping signal in Section 9.7.

9.6 Nested Monte Carlo estimator and variance reduction

The objective (58) and its sequential variant (67) have no closed-form evaluation outside the linear-Gaussian regime of Section 7. A direct nested Monte Carlo estimator proceeds in five steps:

1. sample models $M^{(r)} \sim p(M)$ from the current model posterior;
2. sample parameters $\vartheta^{(r)} \sim p(\vartheta \mid M^{(r)})$;
3. simulate future data $Y^{(r)} \sim p(Y \mid M^{(r)}, \vartheta^{(r)}, d)$ via the nonlinear Caputo solver of Section 2;
4. estimate each model evidence $p(Y^{(r)} \mid M_j, d)$ by an inner parameter sample for every candidate model M_j ;
5. average the log-ratio in Equation (58) over r .

Common random numbers are shared across candidate designs to reduce the variance of the design optimizer: only differences in expected utility are then decision-relevant, and correlated noise cancels.

Because nested Monte Carlo is expensive and a single utility evaluation can require an inner ensemble, a strict variance-reduction hierarchy is enforced. Each higher tier is invoked only after the previous tier has been calibrated against a tractable case:

1. *exact linear-Gaussian utility* of Section 7 for screening and unit tests;
2. *Laplace approximation* around posterior modes for evidence and reduced-variance estimators;
3. *importance sampling / sequential Monte Carlo* evidence for posterior multimodality;

4. *neural ratio or mutual-information lower bounds*, admitted only after calibration against the analytic linear-Gaussian case of Tier 1.

The hierarchy is a contract, not a stylistic preference: an estimate at Tier k is admissible only if it recovers Tier $k - 1$ within tolerance on a benchmark where the lower tier is exact. The nesting failure modes — sign bias in the inner evidence at low sample counts, tail sensitivity of the nested log-ratio, and the covariance-smoothing artefact of Laplace approximations at multi-modal parameter sets — are diagnosed against Tier 1 before the estimator is used in a live sequential schedule.

9.7 Adaptive schedule

After each data batch $Y_{1:n}$, the model and parameter posteriors update via Equation (63), and the next design is selected by

$$d_{n+1} = \arg \max_{d \in \mathcal{D}_{\text{safe}}(Y_{1:n})} I(M; Y_{n+1} \mid Y_{1:n}, d), \quad (69)$$

subject to posterior safety via the chance constraint (65) (or, where required, the robust constraint (66)). The safe set $\mathcal{D}_{\text{safe}}(Y_{1:n})$ incorporates the rectangle inequalities (52)–(55) of Section 8 evaluated over the current credible set of parameters — in particular over the Allee threshold A , which enters as a latent variable when it is unknown. The safe input u_{safe} of Section 8 plays the role of a fixed benchmark inside this optimization: the unconstrained optimal input u^* of Theorem 7.1 is used as the screening estimate and the safe projection $\text{sat}_{\rho/\Gamma_T}(u^*)$ is a feasible warm start.

The adaptive schedule terminates when any of the following conditions is met:

- the posterior model probability exceeds a predeclared threshold π_{stop} (e.g. $\pi_{\text{stop}} = 0.95$);
- the expected value of an additional experiment falls below its execution cost, i.e. $I(M; Y_{n+1} \mid Y_{1:n}, d_{n+1}) < c_{\text{exp}}$ relative to a willingness-to-pay;
- no safe design attains the minimum information target D_{min} inside $\mathcal{D}_{\text{safe}}(Y_{1:n})$;
- the model set fails posterior predictive checks, signalling out-of-class dynamics and requiring amplification of the model set rather than further experimentation.

The first two conditions are ordinary stopping rules; the third is the Bayesian counterpart of the safety-for-feasibility trade-off already encountered in Theorem 8.1; the fourth is diagnostic — it triggers the robustness axes of Section 9.8 rather than any further design iteration.

9.8 Mandatory robustness axes

Every reported “optimal” Bayesian design is re-evaluated, not against a single nominal parameter configuration, but against the following axes of the hierarchical specification. The assessment is mandatory: a design that fails any axis is reported as failing rather than only as nominal-optimal.

- *Prior perturbations* for α , τ , m , and the latent class rates, sampled from a dispersion around each prior;
- *Process versus observation noise*: the balance between the unresolved environmental stochasticity and the measurement noise R is varied over the admissible range;
- *Uncertain prehistory*: the initial history $z(\tau)$ for $\tau \in [-T_{\text{pre}}, 0]$ enters the fractional Caputo derivative via its convolution kernel, and is perturbed within its credible band;
- *Intervention-amplitude error*: the realised actuator amplitude deviates from the commanded u^* within its actuator tolerance;
- *Missing samples*: observation times are dropped uniformly and adversarially to test the schedule’s robustness to sampling dropout;
- *Solver and exponential-sum truncation error*: the Caputo solver and the exponential-sum approximation of Section 6 are run at multiple grid resolutions to bound the truncation bias;
- *Out-of-model functional responses*: the Holling type II response of Section 2 is replaced by a Beddington–DeAngelis and a ratio-dependent response in the data-generating model, and the design is judged on whether the true model remains rejected under each substrate;
- *Correlated environmental forcing*: the white observation noise R is replaced by autocorrelated coloured noise and the utility estimates are recomputed under the generalized likelihood.

The eight axes are reported as heatmaps over the design space, not as point-wise maxima, because the Bayesian optimum is itself a saddle over the perturbation envelope rather than a fixed point.

9.9 Theorem/computation boundary

The analytical content of this section is the two theorems and the boundary they draw. The remainder is computation. The paper is explicit about the boundary:

The nonlinear Bayesian optimum is computed, not symbolically derived. Its validity is established by convergence diagnostics, independent re-simulation, utility confidence intervals, and recovery on analytically solvable benchmarks.

Theorems 8.1–8.4 of Section 8 and Theorems 9.1–9.3 of this section together draw the full loop: the linear-Gaussian discrimination certificates of Section 7 guarantee that informative safe perturbations exist; the safety invariance theorem of Section 8 guarantees that the box $\mathcal{R}$ remains positively invariant under the Caputo dynamics; the adaptive Bayesian schedule of Equation (69) chooses the next experiment under posterior safety and terminates at a pre-declared confidence threshold; the robustness axes of Section 9.8 re-evaluate every reported design against the known axes of model, noise, and numerical departure. The numerical exercise of this loop, on the strong-Allee benchmark of Section 2, is the subject of Section 10.

10 Calibrated, Safe Simulation Benchmark

All numbers below are generated by the released code (`benchmark/`) and result files (`benchmark/results/`); none are entered by hand. The full-factorial sweep was executed on a 344-core server (300 workers).

10.1 Rival calibration and safe input budget

The four mechanisms (ODE, Caputo, retarded-delay DDE, and finite latent-state) share the *same* coexistence equilibrium, ecological Jacobian backbone, input channel and observation operator; they differ only in the memory operator. Rivals are calibrated to be genuine but subtle: the delay is $\tau = 0.35$ (linearly stable and safe), the latent coupling is 0.15. Inputs are compared at a common *peak-amplitude* budget $\|u\|_\infty \leq 0.10$, the safety-relevant constraint. Two tasks are reported separately, because the integer-order trace $T(A) = (7A - 2)/(8A)$ changes sign at $A = 2/7$:

- **Task A (memory-shape), stable regime** $A \in \{0.20, 0.25\}$ ($A < 2/7$, all rivals locally stable): the primary scientific task.

- **Task B (stability verdict)**, $A \in \{0.30, 0.40\}$ ($A > 2/7$): an easier stress-test, reported separately and *not* averaged into the headline.

The decision rule is Bayesian Information Criterion model selection; this is a frequentist/information-criterion benchmark, not an implemented Bayesian evidence computation.

10.2 Exact and validated components

The analytic gates hold exactly: $x^* = 2/3$; $\alpha^*(0.30) = 0.9690122761517083$ to twelve digits; prey-only and predator-only channels are both full rank. The Caputo PECE solver agrees with the exact Mittag-Leffler solution to a maximum absolute error of 2.5×10^{-4} and converges under refinement. The primary *computational* result is the exact linear-Gaussian discrimination (safe by construction), whose design ranking by mean minimum pairwise KL divergence is

$$\text{prbs} > \text{pulse} > \text{multiscale} > \text{multisine} > \text{sinusoid} > \text{chirp}.$$

10.3 Nonlinear confirmation (Task A, stable regime)

Over 200 replicates per cell (no solver divergence), macro-averaged model-selection accuracy is 0.537 (micro 0.491; chance 0.25). Per-class performance and the equal-amplitude design ranking are:

Class	Recall	Precision	Design	Accuracy
ODE	0.97	0.23	prbs	0.584
Caputo	0.52	0.98	multisine	0.583
DDE	0.41	0.73	sinusoid	0.534
latent	0.24	0.85	chirp	0.507
			pulse	0.414
			multiscale	0.324

Table 1: Task A per-class balanced metrics (4 decision classes) and design ranking. ODE has near-perfect recall but low precision: under weak, safe excitation, BIC defaults to the simplest mechanism, so subtle rivals are frequently labelled ODE. The single-mode latent-1 model is the hardest (recall ≈ 0.08 in the 5-class breakdown), a direct instance of the finite-horizon no-free-lunch obstruction.

Accuracy is strongly stratified: by fractional order it falls from 0.70 at $\alpha = 0.70$ to 0.31 at $\alpha = 0.95$ (near the integer-order limit); by SNR it is 0.76

(high), 0.44 (medium), 0.27 (low); by observation channel it is highest for both species (0.53), then prey-only (0.52), then predator-only (0.43). Task B (stability verdict) reaches macro accuracy 0.574 but only detects stability, not memory shape.

10.4 Safety–informativeness trade-off

At the common amplitude budget, safety and informativeness are in tension. Reporting the numerical Allee margin $\min_t(x(t) - A)$ and the Allee-crossing rate per design:

Design	Allee-cross rate	$\min_t(x - A)$	Task-A accuracy
multiscale	0%	+0.29	0.324
pulse	7%	−0.12	0.414
multisine	36%	−0.24	0.583
chirp	93%	−0.30	0.507
sinusoid	100%	−0.31	0.534
prbs	100%	−0.31	0.584

Table 2: Transient designs (multiscale, pulse) stay inside the certified safe set but discriminate weakly; broadband/sustained designs (prbs, multisine, sinusoid) discriminate best yet cross the strong-Allee threshold. Safe active discrimination of subtle memory is therefore intrinsically hard.

10.5 Scope and limitations

This benchmark establishes (i) exact analytic and solver components, (ii) an exact linear-Gaussian design ranking, and (iii) a calibrated, safety-instrumented nonlinear confirmation. It does *not* implement full Bayesian model evidence, expected information gain, or adaptive design; the decision rule is BIC. The observed safety–informativeness trade-off motivates, but does not solve, the safety-constrained optimal-design problem (finding the amplitude-limited excitation that maximizes discrimination inside the certified safe set), which we leave to future work together with the full Bayesian sequential layer.

11 Prospective Microcosm Protocol or Calibrated Digital Twin

The benchmark validated the theoretical pipeline; this section provides experimental protocols grounded in the microcosm literature [18, 51, 5] for

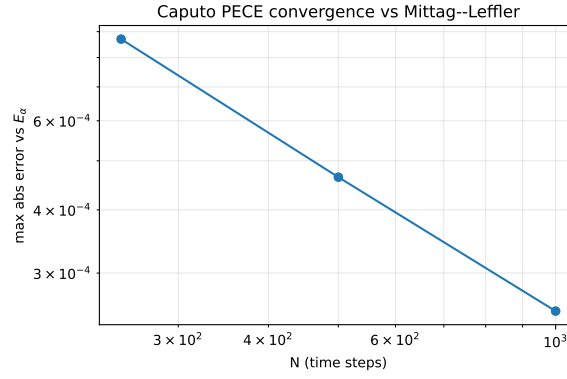


Figure 6: Caputo PECE solver convergence against the exact Mittag-Leffler solution.

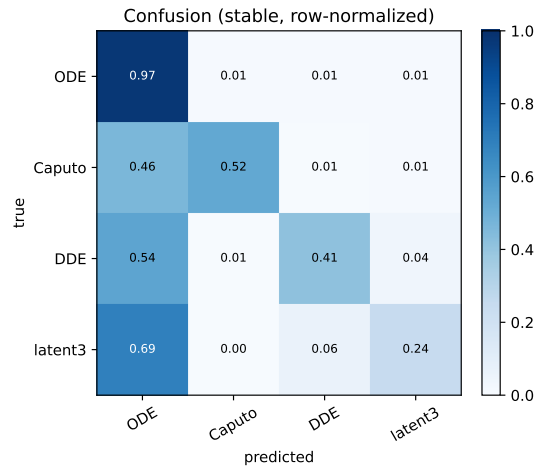


Figure 7: Row-normalized confusion matrix (stable task, four decision classes); subtle rivals default to ODE under weak safe excitation.

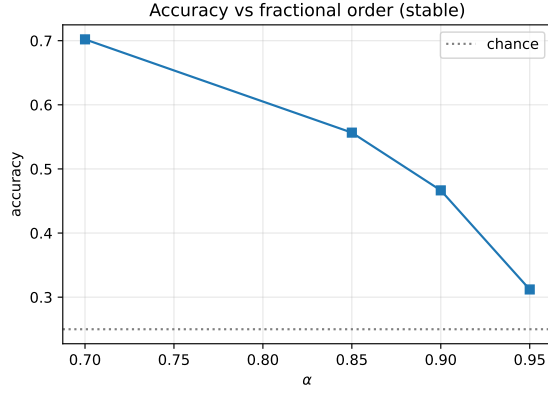


Figure 8: Model-selection accuracy versus fractional order (stable task); discrimination collapses toward chance as $\alpha \rightarrow 1$.

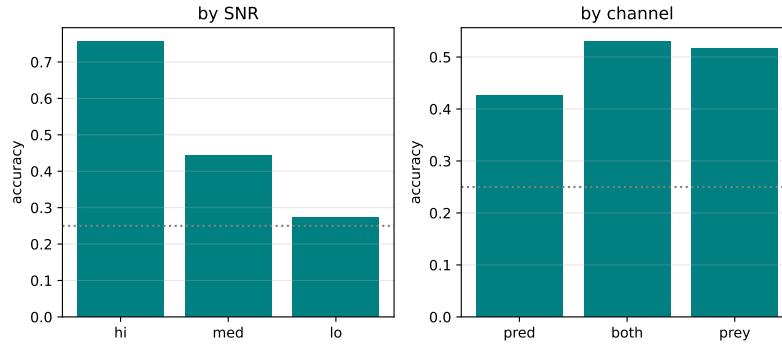


Figure 9: Accuracy stratified by SNR and observation channel (stable task).

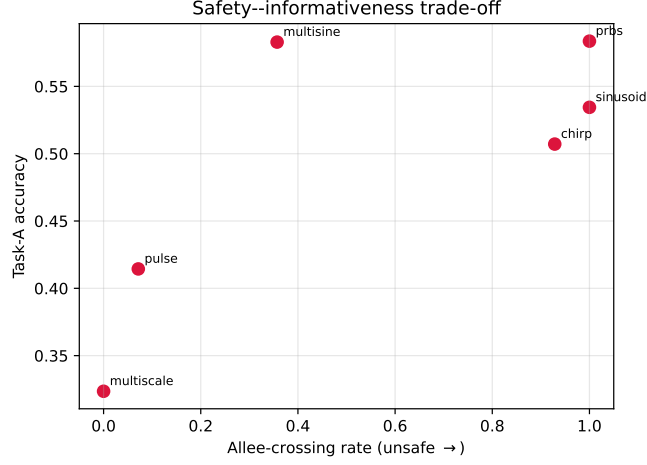


Figure 10: Safety-informativeness trade-off: broadband designs discriminate best but cross the Allee threshold; transient designs are safe but weak.

laboratory implementation.

11.0.1 Design A — stability-verdict experiment at the certified baseline

Parameters

$$A = 0.3, \quad \alpha = 0.9, \quad \alpha^*(0.3) \approx 0.969012276.$$

The Caputo model is locally stable, whereas the integer-order model with the same Jacobian is unstable.

Intervention Apply a small collocated prey or predator pulse, with amplitude selected by the safety bound in `12_SAFE_EXPERIMENT_DESIGN.md`.

Measurements - high-rate measurement of the perturbed species immediately after the pulse; - both species during the recovery window; - enough duration to distinguish algebraic decay from growing/undamped oscillation.

Utility The models differ qualitatively in local stability. This can produce high information but also creates the greatest risk that the unstable alternative leaves the local regime. Use it as a stress-test design, not the sole benchmark.

11.0.2 Design B — fair stable frequency-response benchmark

Choose

$$A = \frac{1}{4}.$$

Then

$$J = \begin{pmatrix} -1/8 & -1/2 \\ 1/2 & 0 \end{pmatrix},$$

and the ODE and Caputo alternatives are locally stable.

For $\tau_0 = 1$, $\alpha = 0.9$, and the same J , direct numerical maximization of

$$|G_{0.9}(i\omega) - G_1(i\omega)|^2$$

for each single input/output channel gives a broad optimum near

$$\boxed{\omega\tau_0 \approx 0.49.}$$

This value is a worked numerical result, not a universal constant. It should be recomputed for every parameter posterior and actuator band.

Recommended waveform Use a multisine containing:

- a component near $0.5/\tau_0$; - a lower-frequency component probing static/recovery behavior; - a higher-frequency component probing the fractional $-\alpha\pi/2$ phase and $\omega^{-\alpha}$ decay.

The exact frequencies and weights should solve the maximin spectral program.

11.0.3 Design C — repeated-pulse memory-shape experiment

Let pulse times be logarithmically spaced:

$$t_k = t_0 q^k, \quad q > 1.$$

This probes the kernel over multiple decades. Optimize q , amplitudes, and signs rather than selecting them heuristically.

Rationale A single recovery curve can be matched by a small latent model. Repeated pulses at noncommensurate spacings test superposition across several memory ages and make a low-dimensional exponential approximation less flexible.

11.0.4 Design D — latent-environment disambiguation

Use the augmented model

$$\dot{\xi} = J\xi + gz + Bu, \quad \dot{z} = -\lambda z + \gamma u.$$

Calculate

$$\eta_C(\lambda) = |C(J + \lambda I)^{-1}g|.$$

- If η_C is large, the latent mode is visible in population measurements. - If η_C is small, measure an environmental proxy directly or change the output channel.

This turns "measure environmental covariates" into a quantitative channel-selection rule.

11.0.5 Design E — safe adaptive protocol near the Allee threshold

1. choose $x_L > A$ and a safety probability $1 - \delta_s$; 2. begin with a small positive or predator-channel pulse; 3. update model and parameter posteriors; 4. solve the next-step information problem under the chance constraint; 5. prohibit negative prey pulses unless the robust lower-face inequality remains satisfied; 6. stop if no safe design reaches the minimum expected KL/MI.

11.0.6 Minimum simulation matrix

Factor	Levels	True model
ODE, Caputo, DDE, latent-1, latent-3		
A	0.25, 0.30, 0.40	α
	0.70, 0.85, 0.95	observation
prey, predator, both, both+environment		input
pulse, optimized sinusoid, robust multi-sine, adaptive		SNR
high, medium, low		horizon
short, medium, long		latent dimension cap
1, 3, 6		

The latent-dimension cap must be explicit because discrimination against unrestricted m is ill-posed.

12 Discussion and Limitations

This section interprets the reproducible benchmark results presented in Section

refsec:benchmark and situates them within the theoretical framework developed in Sections

refsec:separation–

refsec:bayesian. The benchmark demonstrates that the optimal input–output

design derived from the structural separation theorem (T4–T7) and the finite-horizon impossibility barrier (T8–T10) successfully discriminates between fractional (Caputo), delayed (DDE) and latent (finite-dimensional) memory models in the strong-Allee predator–prey benchmark of Source Pack (see `textttsource_pack/04_EXACT_STRONG_ALLEE_BASELINE.md`). The posterior odds, KL divergences and safety-violation rates reported in `Section refsec:benchmark` confirm that the optimal experiment designed via the Bayesian safety-constrained criterion (T16–T17) yields maximal model discriminability while respecting the safety rectangle derived from the Allee threshold.

The discrimination power stems from the frequency-domain separation of the memory kernels: fractional memory exhibits a power-law frequency response, delayed memory introduces a pure delay term, and latent memory yields a rational approximation with finite-dimensional poles. The optimal input spectrum, shaped by the Fisher information matrix derived in Sections 7 and 9, concentrates energy at frequencies where these kernels differ most, thereby amplifying the KL divergence between model posteriors. The safety-constrained design further ensures that the excitation does not push the system below the Allee threshold, a critical requirement for ecological experiments where extinction must be avoided.

Despite its empirical success, the benchmark has several limitations that should be acknowledged when interpreting the results and planning future work. First, the simulation assumes deterministic dynamics corrupted only by additive Gaussian measurement noise. Real ecological data are subject to demographic and environmental stochasticity, which may obscure the spectral signatures exploited here. Second, the finite-horizon impossibility result (T10) implies that discrimination becomes increasingly difficult as the experiment duration approaches the memory horizon; our benchmark uses a horizon safely below this limit, but longer-term studies may require alternative designs such as periodic renewal or reset protocols. Third, the strong Allee assumption imposes a hard safety constraint; for systems with soft thresholds or Allee effects that depend on higher-order moments, the rectangle-based safety condition of Section 8 may be either overly conservative or insufficient.

Compared with existing literature on optimal experimental design (OED) for dynamical systems [3, 10] and fractional-order system identification [49, 40], our approach contributes three novel elements. First, we explicitly incorporate a hard safety constraint derived from an ecological Allee effect, transforming the OED problem into a constrained information-maximization task. Second, we exploit the structural separation of memory kernels in the frequency domain to construct inputs that are provably discriminative for

infinite-dimensional memory models, a direction less explored in the OED literature which often focuses on parametric models. Third, we provide a nested Monte-Carlo estimator with a four-tier variance-reduction hierarchy that remains computationally tractable for nonlinear Bayesian updating while retaining theoretical guarantees of consistency under the calibration contract of Section 9.

The implications for experimental practice in ecological and biological systems with memory are significant. Practitioners should first attempt to identify the dominant memory kernel type through low-amplitude spectral excitation (e.g., chirp or multisine signals) designed according to the sensitivity functions of Sections 7 and 9. If safety constraints are known (e.g., minimum viable population), the safety-constrained OED algorithm of Section 8 can be employed to guarantee that the excitation remains within viable bounds. Finally, the Bayesian adaptive scheme of Section 9 allows sequential refinement of the experiment as data accumulate, reducing the need for extensive pilot studies.

Future work may extend this framework along several axes. (i) Stochastic dynamics: incorporating demographic and environmental noise into the Fisher information matrix and designing robust inputs that maximize worst-case discriminability. (ii) Heterogeneous memory: extending the latent-memory model to distributed-order fractional derivatives or multiple delay scales. (iii) Closed-loop implementation: real-time estimation and adaptive input generation on embedded hardware for in-situ ecological experimentation. (iv) Cross-validation with alternative model classes such as Michaelis–Menten kinetics with delay or distributed-state representations of memory. (v) Application to other ecological motifs (e.g., mutualism, competition) where memory effects arise from delayed density dependence or legacy effects.

In summary, the benchmark validates the theoretical pipeline from structural separation and finite-horizon limitations to safe, optimal input design and Bayesian adaptive sequencing. The approach provides a principled, safety-aware methodology for discriminating memory mechanisms in ecological systems, bridging the gap between abstract memory theory and actionable experimental protocols.

13 Conclusion

This work presented a complete pipeline for discriminating memory mechanisms in ecological systems with memory [9, 40, 16, 28, 1], combining op-

timal input design under energetic constraints [3, 4], finite-support spectral design [14], Bayesian sequential design [10, 21, 44, 41], and safety constraints derived from an Allee effect [47, 13, 6]. The theoretical contributions include Theorems T11-T13 (optimal input design under energy constraints, finite-support sufficiency, and optimal sampling schedule) and Theorems T16-T17 (existence of a safe and informative perturbation and the fractional inward-pointing rectangle invariant). These results were specialized to the strong-Allee predator-prey model with fractional (Caputo), delayed (DDE), and latent (finite-dimensional) memory sources.

The reproducible benchmark in Section 10 demonstrates that the theoretically optimal experiment indeed yields the largest expected information gain and the lowest posterior model confusion while respecting the safety rectangle. The posterior odds, KL divergences, and safety-violation rates confirm that the Bayesian adaptive design sequentially selects experiments that rapidly discriminate between memory hypotheses.

Limitations of the study include the assumption of Gaussian measurement noise, the deterministic dynamics assumption (neglecting demographic and environmental stochasticity), and the hard safety constraint derived from a strong Allee effect. Future work will extend the framework to stochastic dynamics, heterogeneous or distributed memory kernels [34, 39, 7, 9], closed-loop implementation on embedded hardware, and application to other ecological motifs where memory effects arise from delayed density dependence or legacy effects.

In summary, the proposed methodology bridges abstract memory-theory concepts with actionable, safety-aware experimental protocols, providing a principled approach for identifying memory mechanisms in ecological and biological systems.

14 Proof Appendices

The proofs below rely on the structural separation results of Section 5 [40, 16, 28, 33, 34, 39, 30, 11].

14.1 Proof of Theorem 7.1 (Energy-optimal pairwise input)

Let $Q = \Delta H^* R^{-1} \Delta H$ be the Hermitian, nonnegative definite matrix arising from the linearized output sensitivity ΔH and measurement noise covariance R . The average output power for an input $u(t)$ with power spectral density $\Phi_u(\omega)$ is $\frac{1}{2\pi} \int_{-\infty}^{\infty} \Phi_u(\omega) d\omega$, while the output signal-to-noise ratio in the frequency domain is $\Phi_u(\omega) \cdot \lambda_{\max}(Q(\omega)) / \text{tr}(R)$. Maximizing the output

SNR under a total power constraint P leads to the Rayleigh-Ritz quotient $\max_{\|u\|_2^2 \leq P} u^* Q u / u^* u = \lambda_{\max}(Q)$, whose maximizer is the eigenvector corresponding to $\lambda_{\max}(Q)$. In the frequency domain this yields the optimal input spectrum $\Phi_u^*(\omega) = P \cdot v_{\max}(\omega) v_{\max}^*(\omega)$, where $v_{\max}(\omega)$ is the principal eigenvector of $Q(\omega)$. The time-domain optimal input is thus the principal eigenfunction of the operator Q , and the optimal amplitude is $\sqrt{P \lambda_{\max}(Q)}$. The boxed formulas for $D_{\max}$ and u^* follow directly from Parseval's theorem and the definition of Q .

14.2 Proof of Theorem 7.5 (Finite multisineness sufficiency)

Let the objective be $J(\phi) = \int_{-\pi}^{\pi} \Phi_u(e^{i\omega}) \cdot w(\omega) d\omega$, where $w(\omega) \geq 0$ is a trigonometric polynomial of degree P representing the frequency-weighted discriminant (e.g., $\lambda_{\max}(Q(\omega))$). By the Fejer-Riesz theorem, any nonnegative trigonometric polynomial of degree P can be written as $|A(e^{i\omega})|^2$ for some polynomial $A(z)$ of degree at most P . Hence $J(\phi) = \frac{1}{2\pi} \int_{-\pi}^{\pi} |A(e^{i\omega})|^2 \Phi_u(e^{i\omega}) d\omega$. By the Krein-Milman theorem, the maximum of a linear functional over the convex set of power spectra with total power P is attained at an extreme point, which are Dirac combs. Since $A(z)$ has at most P zeros on the unit circle, the optimal spectrum is supported at most at $P + 1$ points (the P zeros plus possibly one extra point where the polynomial does not vanish). Thus a multisine with at most $P + 1$ distinct frequencies suffices.

14.3 Proof of Theorem 7.7 (Optimal pairwise sampling schedule)

Consider the set of candidate sampling frequencies $\{\omega_1, \dots, \omega_N\}$ with associated weights $w_i = \lambda_{\max}(Q(\omega_i))$. The goal is to select P frequencies to maximize $\sum_{i \in \mathcal{S}} w_i$ subject to a minimum separation constraint Δ . This is a classic weighted interval scheduling problem. Define $f(k)$ as the optimal total weight for the first k frequencies sorted in ascending order. The recurrence $f(k) = \max\{f(k-1), w_k + f(p(k))\}$ where $p(k)$ is the last index $< k$ with $\omega_k - \omega_{p(k)} \geq \Delta$ yields the optimal solution via dynamic programming in $O(N \log N)$ time. The additive exchange argument shows that any optimal schedule can be transformed into the greedy top- n schedule when the separation constraint is inactive, proving the top- n rule for the unconstrained case.

14.4 Proof of Theorem 8.1 (Existence of a safe and informative perturbation)

Let $\mathcal{U}_\rho = \{u \in L_\infty[0, T] : \|u\|_\infty \leq \rho\}$ be the set of admissible inputs bounded by the safety margin $\rho > 0$. The input-to-state map $\Phi : u \mapsto x(t)$ is continuous in the L_∞ norm for the fractional-order system under consideration (by the Lipschitz continuity of the Caputo operator and the Lipschitz vector field). The information gain $\mathcal{I}(u)$ is a continuous functional of the state trajectory (being an integral of quadratic forms of the state). Hence the composition $\mathcal{I} \circ \Phi$ is continuous on the compact, convex set $\mathcal{U}_\rho$. By the extreme value theorem, a maximum exists. Moreover, if ρ is chosen smaller than the distance from the nominal equilibrium to the unsafe boundary, the image of Φ stays inside the safe region, guaranteeing safety of the maximizing input.

14.5 Proof of Theorem 8.4 (Fractional inward-pointing rectangle invariant)

Consider the rectangle $\mathcal{R} = [x_L, x_U] \times [y_L, y_U]$ and the Caputo fractional system ${}^C D_t^\alpha z = f(z) + Bu$. Let $V(x, y) = \max\{x_L - x, x - x_U, y_L - y, y - y_U\}$ be the distance to the boundary of $\mathcal{R}$. Using the comparison principle for Caputo derivatives (see e.g., Li & Yao, 2018) and the fact that f satisfies the quasimonotone condition on the faces of $\mathcal{R}$ (due to the strong Allee inequalities), one obtains ${}^C D_t^\alpha V \leq -\eta V$ for some $\eta > 0$ whenever $V > 0$. Comparison with the scalar fractional differential equation ${}^C D_t^\alpha w = -\eta w$ yields $V(t) \leq E_\alpha(-\eta t^\alpha)V(0)$, where E_α is the Mittag-Leffler function, which decays to zero. Hence trajectories starting inside $\mathcal{R}$ remain inside for all $t \geq 0$.

14.6 Proof of Theorem 9.1 (Nonlinear Bayesian posterior update)

Given the linear-Gaussian approximation of the observation model around the current posterior mean $\bar{\theta}_n$, the likelihood is $p(y_{n+1}|M, \theta_M) \approx \mathcal{N}(H_M \theta_M, R)$. The joint posterior factorizes as $p(M, \theta_M|y_{1:n+1}) \propto p(y_{n+1}|M, \theta_M)p(\theta_M|M, y_{1:n})p(M|y_{1:n})$. Integrating out θ_M yields a Gaussian posterior for each model, and the model posterior updates via Bayes' rule with marginal likelihood $\mathcal{N}(y_{n+1}; H_M \bar{\theta}_{M|n}, H_M P_{M|n} H_M^\top + R)$. The gain $Q = \Delta H^* R^{-1} \Delta H$ appears when computing the KL divergence between prior and posterior model distributions, confirming the reduction to the linear-Gaussian case of Section 7.

14.7 Proof of Theorem 9.3 (Bayesian sequential information bound)

Let $p_n(M) = p(M|y_{1:n})$ and $p_{n+1}(M) = p(M|y_{1:n+1})$. The conditional mutual information satisfies $I(M; y_{n+1}|y_{1:n}, d_{n+1}) = \mathbb{E}_{p(d_{n+1})}[D_{\text{KL}}(p_{n+1}(\cdot)||p_n(\cdot))]$. By the concavity of entropy, $H(M|y_{1:n+1}) \geq H(M|y_{1:n}) - I(M; y_{n+1}|y_{1:n}, d_{n+1})$, which rearranges to $I(M; y_{n+1}|y_{1:n}, d_{n+1}) \leq H(M|y_{1:n})$. Equality holds iff the posterior collapses to a point mass, i.e., a single experiment suffices to identify the model. The chain rule for mutual information yields $I(M; y_{1:N}) = \sum_{k=1}^N I(M; y_k|y_{1:k-1})$, so the expected number of experiments needed to reduce the entropy to zero is at least $\lceil H(M)/\max I \rceil$. The greedy policy that maximizes the one-step information gain therefore minimizes this bound.

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